

# Reply to the Referee Report

JHEP\_282P\_0726: *Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order*

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**How this document is organised.** Part I answers the referee’s four comments one by one. Each answer quotes the comment, gives the reply, and lists the changes to make in the manuscript. For Comment 2 two alternative replies are given (Version A with a regularisation prescription, Version B without); send one of them. Part II lists further errors found in a line-by-line check of the manuscript. Each entry names the equation, what is printed, the correction, and how it was checked. Part III contains the complete step-by-step calculations behind the replies to Comments 1 and 2. It is written for the author and should not be sent. Equation numbers refer to the submitted version (v1). Orange boxes are notes to the author and should be removed before the reply is sent.

## Note to the author (not for the referee)

Two kinds of statement appear below. *Derived*: re-derived here from the paper’s own matrix elements (App. D) and state normalisations (App. B), or checked by dimensions. *To verify*: plausible but depends on conventions that could not be fully reconstructed from the PDF. Every *To verify* item should be checked against your own notes before you submit. The new  $t$ -channel expression in Part I is written in terms of the paper’s matrix elements. Its final  $(2\pi)$  factors and colour/argument labels must be fixed in the conventions of Eq. (5.7).

Dear Editor,

I thank the referee for a careful reading of the manuscript and for comments that improve the paper substantially. Below I reply to each comment and describe the changes made. In the course of the revision I also re-checked the whole manuscript and corrected a number of further errors, listed in Part II.

## Part I — Reply to the referee

### Comment 1: the $t$ -channel gluon exchange in Sec. 4.3.1

#### Referee, comment 1

It seems that in Sec 4.3.1. one is neglecting a “ $t$ -channel gluon exchange” contribution, where one passes through a three-gluon intermediate state. This is a significant part of the result. It is also curious that the authors do not notice the absence of this contribution in the unitarity relations, which should not hold when it is neglected.

**Reply.** The referee is correct. Eq. (4.82) contains all three colour structures,  $f^{a'ad}f^{a'cb}$  ( $t$ -channel,  $k \rightarrow r$ ),  $f^{a'ac}f^{a'db}$  ( $u$ -channel,  $k \rightarrow q$ ) and  $f^{a'ab}f^{a'dc}$  ( $s$ -channel), and each of them carries two pieces:

- the four-gluon contact interaction (the  $\delta\delta$  terms, from  $F_{ij}F_{ij}$ ), a pointlike vertex in which no gluon is exchanged;
- the *instantaneous* gluon exchange in that channel (the terms with plus-momentum ratios, from  $\Pi^2$ ), with denominators  $-(k^+ - r^+)^2$ ,  $-(k^+ - q^+)^2$  and  $(k^+ + p^+)^2$ , respectively.

In light-cone perturbation theory a gluon exchange consists of this instantaneous part and a *propagating* part, which is second order in  $H_{ggg}$  and passes through an intermediate state. This

follows from the light-cone-gauge propagator,

$$\frac{d^{\mu\nu}(k)}{k^2 + i0} = \sum_{\lambda=1,2} \frac{\epsilon_\lambda^\mu \epsilon_\lambda^{\nu*}}{k^2 + i0} + \frac{n^\mu n^\nu}{(k^+)^2},$$

whose first term propagates in light-cone time and whose second term is instantaneous. For the  $s$ -channel the paper includes both parts: the instantaneous part in Eq. (4.82) and the propagating part, with a one-gluon intermediate state, in Eq. (4.84). For the  $t$ - and  $u$ -channels only the instantaneous part was included. The propagating part was missed: one incoming gluon emits a gluon through  $H_{ggg}$  and the other incoming gluon absorbs it through a second  $H_{ggg}$ , so the intermediate state has three gluons. This is the contribution the referee refers to. It is evaluated below and added to the paper.

*The missing diagram.* Label the incoming gluons  $g_i^a(k)$ ,  $g_j^b(p)$ , the outgoing gluons  $g_l^d(r)$ ,  $g_n^c(q)$  (the paper writes the polarisation of  $q$  as  $k$ ; we use  $n$  to avoid confusion with the momentum  $k$ ), and the exchanged gluon  $g_m^e(r')$ , with  $k + p = r + q$ . A gluon with positive plus momentum always propagates forward in light-cone time, so two time orderings are possible, and for given external momenta exactly one of them contributes:

- $k^+ > r^+$ :  $k \rightarrow r + r'$  first, then  $r' + p \rightarrow q$ . The intermediate state is  $|g(r)g(r')g(p)\rangle$  with  $r' = k - r$ .
- $k^+ < r^+$ :  $p \rightarrow q + r''$  first, then  $r'' + k \rightarrow r$ . The intermediate state is  $|g(k)g(r'')g(q)\rangle$  with  $r'' = p - q$ .

The two orderings are shown in Fig. 1(a),(b), together with the  $s$ -channel and instantaneous exchanges already in the paper, (c),(d).

In LCPT the  $t$ - and  $u$ -channel exchanges give

$$|(\Psi_{ji}^{ba}(k,p))_{gg}^t\rangle = \int d^3r d^3q |g_l^d(r)g_n^c(q)\rangle [T^{(a)}(r,q) + T^{(b)}(r,q)], \quad (1)$$

$$\begin{aligned} T^{(a)} &= \frac{\Theta(k^+ - r^+)}{E_{gg}(k,p) - E_{gg}(r,q)} \int d^3r' \frac{\langle g_n^c(q) | H_{ggg} | g_m^e(r') g_j^b(p) \rangle \langle g_l^d(r) g_m^e(r') | H_{ggg} | g_i^a(k) \rangle}{E_g(k) - E_{gg}(r,r')}, \\ T^{(b)} &= T^{(a)} \Big|_{(k,a,i) \leftrightarrow (p,b,j), (r,d,l) \leftrightarrow (q,c,n)}. \end{aligned} \quad (2)$$

The  $u$ -channel is included through the symmetry of the two-gluon state: with  $T = T^{(a)} + T^{(b)}$ , Eq. (1) equals

$$\frac{1}{2} \int d^3r d^3q |g(r)g(q)\rangle [T(r,q) + T(q,r)],$$

where  $T(q,r)$  is the  $u$ -channel. The spectator's energy cancels in the intermediate denominator:

$$E_{gg}(k,p) - E_{ggg}(r,r',p) = E_g(k) - E_{gg}(r,r').$$

Inserting the vertices Eq. (D.3) and doing the  $r'$  integral gives

$$T^{(a)} = -\frac{g^2}{32\pi^3} f^{ead} f^{ecb} \frac{\delta^{(3)}(k + p - r - q) \Theta(k^+ - r^+)}{\sqrt{k^+ p^+ q^+ r^+}} \frac{\xi P_{ilnj}}{\kappa^2 [E_{gg}(k,p) - E_{gg}(r,q)]}, \quad (3)$$

where  $\xi = r^+/k^+$ ,  $\kappa = \mathbf{r} - \xi \mathbf{k}$  is the relative transverse momentum of the splitting  $k \rightarrow r + r'$ , and  $P_{ilnj} = \sum_m V_{iml}(k; r', r) V_{nmj}(q; r', p)$  is the product of the two polarisation brackets of Eq. (D.3) (its explicit form is given in the revised manuscript). The colour factor  $f^{ead} f^{ecb}$  is the  $t$ -channel structure  $f^{a'ad} f^{a'cb}$  of (4.82). The intermediate denominator,  $E_g(k) - E_{gg}(r,r') = -\kappa^2/[2k^+ \xi(1-\xi)]$ , vanishes only at the collinear endpoint, where  $T^{(a)} \sim 1/|\kappa|$  is integrable.

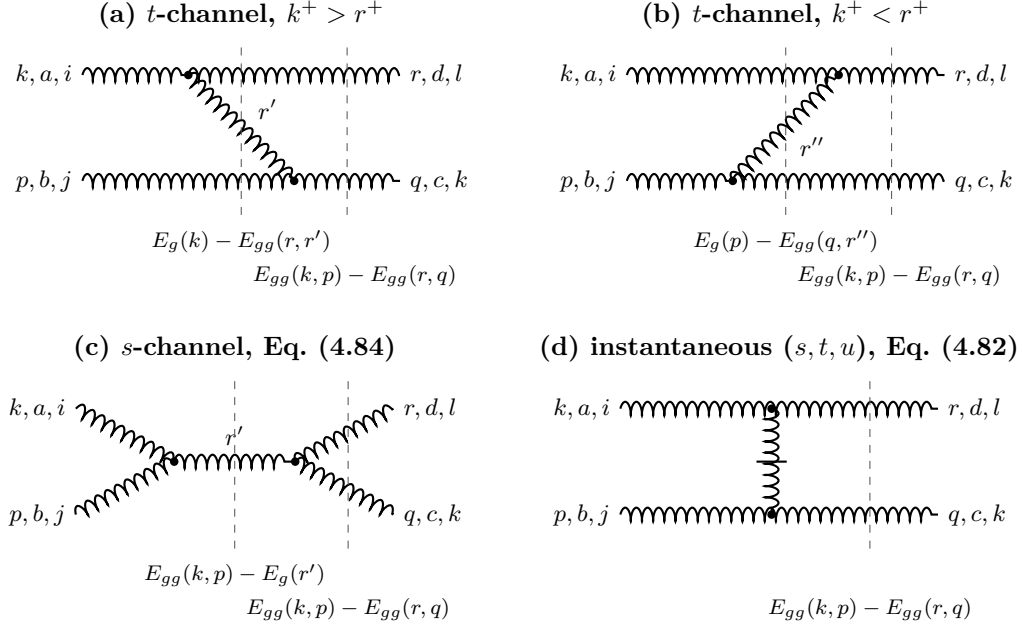


Figure 1: Gluon exchange between two incoming soft gluons at  $\mathcal{O}(g^2)$ . Light-cone time runs from left to right; dashed vertical lines mark the intermediate and final states, with the corresponding energy denominators written below. Diagrams (a) and (b) are the two light-cone time orderings of the non-instantaneous  $t$ -channel exchange, which are missing from Sec. 4.3.1; their intermediate state has *three* gluons. Plus-momentum positivity of the exchanged gluon ( $r'^+ = k^+ - r^+ > 0$  in (a),  $r''^+ = r^+ - k^+ > 0$  in (b)) means that exactly one of them contributes for given external momenta. The  $u$ -channel diagrams follow from  $r \leftrightarrow q$ . For comparison, (c) is the  $s$ -channel exchange already contained in Eq. (4.84), and (d) is the instantaneous gluon exchange (crossed line). Eq. (4.82) contains the instantaneous exchange in all three channels, one per colour structure, but not the propagating  $t$ - and  $u$ -channel exchanges (a), (b).

*Expression in terms of the  $\Omega$  coefficients.* Write  $\omega_1(f \leftarrow i) = V_{fi}/(E_i - E_f)$  for the first-order LCPT amplitude and use  $V_{fm} = \omega_1(f \leftarrow m)(E_m - E_f)$ . Every second-order amplitude with a single intermediate state  $m$  can then be written as

$$\omega_2(f \leftarrow i) = \sum_m \omega_1(f \leftarrow m) \omega_1(m \leftarrow i) \frac{E_m - E_f}{E_i - E_f}. \quad (4)$$

From Eq. (5.5),  $\langle gg | H_{ggg} | g \rangle = 2(2\pi)^{-9/2} (E_{\text{parent}} - E_{\text{daughters}}) \mathbb{C}$ , so the new contribution to  $\mathbb{D}_2$  is

$$\begin{aligned} \mathbb{D}_{2\ln ji}^{(t)dcba}(k, p, q, r) &= 2 \int \frac{d^3 r'}{(2\pi)^3} \mathbb{C}_{jmn}^{bec\dagger}(q, r', p) \mathbb{C}_{lmi}^{dea}(k, r', r) \frac{E_g(q) - E_g(r') - E_g(p)}{E_{gg}(k, p) - E_{gg}(r, q)} \\ &\quad + (\text{ordering (b)}), \end{aligned} \quad (5)$$

where ordering (b) follows from the relabelling in Eq. (2), and only one of the two terms is non-zero for given momenta. The  $u$ -channel enters through the term  $\mathbb{D}_2^{dcab}(p, k, q, r)$  of Eq. (4.17). The factor 2, compared with the  $s$ -channel term of (5.7), is bookkeeping: the  $s$ -channel amplitude is symmetric under  $r \leftrightarrow q$  and is shared between  $\mathbb{D}_2^{dcba}(k, p, q, r)$  and  $\mathbb{D}_2^{dcab}(p, k, q, r)$ , whereas the  $t$ - and  $u$ -channel amplitudes are separate functions. This term is  $\rho$ -independent, so the ordering of  $\mathbb{C}$  and  $\mathbb{C}^\dagger$  does not matter here.

*Why unitarity signals the omission.* The referee is right that the unitarity relation Eq. (4.11) cannot hold without this term. From Eq. (4) and its Hermitian conjugate with  $i \leftrightarrow f$ , for each intermediate

state  $m$  one finds

$$\begin{aligned}\omega_2(f \leftarrow i) + [\omega_2(i \leftarrow f)]^\dagger &= \frac{V_{fm}V_{mi}}{E_i - E_f} \left[ \frac{1}{E_i - E_m} - \frac{1}{E_f - E_m} \right] = -\frac{V_{fm}V_{mi}}{(E_i - E_m)(E_f - E_m)} \\ &= -[\omega_1(m \leftarrow f)]^\dagger \omega_1(m \leftarrow i).\end{aligned}\tag{6}$$

This is the  $\mathcal{O}(g^2)$  unitarity condition Eq. (4.5),  $\Omega^{(2)} + \Omega^{(2)\dagger} = -\Omega^{(1)\dagger}\Omega^{(1)}$ , and it holds *intermediate state by intermediate state*. The ordering  $V_{fm}V_{mi}$  is kept throughout, so the identity also holds for the non-commuting,  $\rho$ -dependent coefficients. For the intermediate state  $|g(r)g(r')g(p)\rangle$  of Fig. 1(a),  $\omega_1(m \leftarrow i)$  is the splitting  $\mathbb{C}(k; r', r)$  and  $\omega_1(m \leftarrow f)$  is the splitting  $\mathbb{C}(q; r', p)$ , so the right-hand side is the product  $-\mathbb{C}^\dagger(q; r', p) \mathbb{C}(k; r', r)$ , in which the internal momentum  $r'$  joins the incoming line  $k$  to the outgoing line  $q$ . Products of this “ $t$ -type” appear on the right-hand side of Eq. (4.11). By Eq. (6) they can only be matched by the Hermitian part of a second-order diagram with the same intermediate state, which is exactly the  $t$ -channel diagram. The  $\mathbb{D}_2$  of Eq. (5.7) contains no such term, so it cannot satisfy Eq. (4.11).

In the submitted version Eq. (4.11) was not explicitly checked against Eq. (5.7); this was my oversight. In the revised version the check is done term by term using Eq. (6), and it is satisfied once the  $t$ - and  $u$ -channel terms are included.

*Consequences elsewhere.* The same exchange appears as a disconnected piece (third gluon as spectator) in the three-gluon sector Eq. (4.18). It is accounted for automatically because the normal-ordered operator  $\mathbb{D}_2 a^\dagger a^\dagger a a$  acts on every pair. In Sec. 6, the new term in  $\mathbb{D}_2$  is needed to cancel the part of  $\frac{1}{2}[H_{ggg}, \Omega^{(1)}]$  with  $t$ -channel structure in the  $2 \rightarrow 2$  sector (see also Part II, item A3).

#### Proposed changes to the manuscript

1. Sec. 4.3.1: after Eq. (4.82), state that each of its three colour structures contains the contact term and the *instantaneous* exchange in the  $t$ -,  $u$ - or  $s$ -channel. Add the propagating  $t$ - and  $u$ -channel exchange diagrams (with figure), Eqs. (1)–(3), with the explicit polarisation tensor  $P_{ilnj}$ . Change “six diagrams” accordingly.
2. Eq. (4.95): add  $|\Psi_{ji}^{ba}(k, p)\rangle_{gg}^t$ . Eq. (5.7): add  $\mathbb{D}_2^{(t)}$  of Eq. (5).
3. Sec. 4 (after Eq. (4.14)) and Sec. 5: state that  $\mathbb{B}_3$  and  $\mathbb{D}_2$  are fully fixed by the matching. The unitarity relations (4.10), (4.11) are then *checks*, not determinations. Add the general identity Eq. (6) and state explicitly that the checks are satisfied.
4. Correct the sentence after Eq. (5.12) and the list on p. 14 (“The coefficients  $\mathbb{A}_1, \dots$  are determined by matching”) so that they agree with item 3.

#### Note to the author (not for the referee)

*Your question:* “(4.82) has both instantaneous and usual terms; the usual terms have all three channels.” The usual (non-instantaneous) terms of (4.82) do carry all three colour structures, but they are the contact four-gluon vertex, not gluon exchanges. The covariant four-gluon vertex has the same three colour structures and is still a single diagram. A  $t$ -channel exchange needs a gluon that propagates between two  $H_{ggg}$  vertices. That term is  $g \times g$ , has a second energy denominator, and is a function of the transverse momenta in its numerator, none of which is true of (4.82). Part III.1 gives the full argument, the propagator decomposition, and the step-by-step calculation of Eqs. (1)–(5), including the explicit  $P_{ilnj}$ . It also contains a cross-check showing that the  $s$ -channel term of (5.7) does not follow from (4.84) as printed (Step 13).

## Comment 2: vanishing $2g \rightarrow 2g$ energy denominator

### Referee, comment 2

This is related to both the “ $t$ -channel gluon exchange” contribution mentioned above and the contribution already included in (4.82). The  $2g \rightarrow 2g$  energy denominator can go to zero and thus needs to be regularized. How this is done should be discussed in the manuscript.

### Note to the author (not for the referee)

Two alternative replies follow. **Version A** introduces the prescription; **Version B** argues that the results of the paper do not need one. Both begin with the same common part, which should be sent with either version. The detailed calculations behind every statement are in Part III.2, which also explains which version is safer.

**Common part (for both versions).** **Reply.** I thank the referee. The overall energy denominator can indeed vanish.

*Why the denominator can vanish.* The  $\delta$ -function in Eq. (4.82) conserves the plus and transverse momenta, but not the light-cone energy; non-conservation of  $p^-$  at the vertices is what produces energy denominators in LCPT. After using it,  $q = k + p - r$ , the denominator is still a function of the free outgoing momentum  $r$ :

$$D(r) = \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+} - \frac{\mathbf{r}^2}{2r^+} - \frac{(\mathbf{k} + \mathbf{p} - \mathbf{r})^2}{2(k^+ + p^+ - r^+)}.$$

This vanishes at  $r = p$  (so  $q = k$ ), which lies inside the integration region, while the numerator of Eq. (4.82) stays finite there. More generally, writing  $E_{gg}(k, p) = \mathbf{P}^2/2P^+ + \kappa^2/[2P^+\xi(1-\xi)]$  with  $\xi = k^+/P^+$  and  $\kappa = (1-\xi)\mathbf{k} - \xi\mathbf{p}$ , the denominator vanishes on the whole surface

$$\frac{\kappa^2}{\xi(1-\xi)} = \frac{\kappa'^2}{\xi'(1-\xi')},$$

see Fig. 2(a),(c). This cannot happen for the vacuum incoming state of Sec. 4.1, where  $E_i = 0 < E_f$ , which is why the issue first appears with incoming soft gluons.

*Which coefficients keep the pole.* When a transition consists of two independent sub-processes (for instance, one gluon splits while another is absorbed by the valence), both time orderings contribute. If the sub-processes change the free energy by  $a$  and  $b$ , so that  $E_i - E_f = a + b$ , then

$$\frac{1}{(a+b)a} + \frac{1}{(a+b)b} = \frac{1}{ab},$$

and the overall denominator cancels in the sum. A pole therefore remains only for connected transitions:

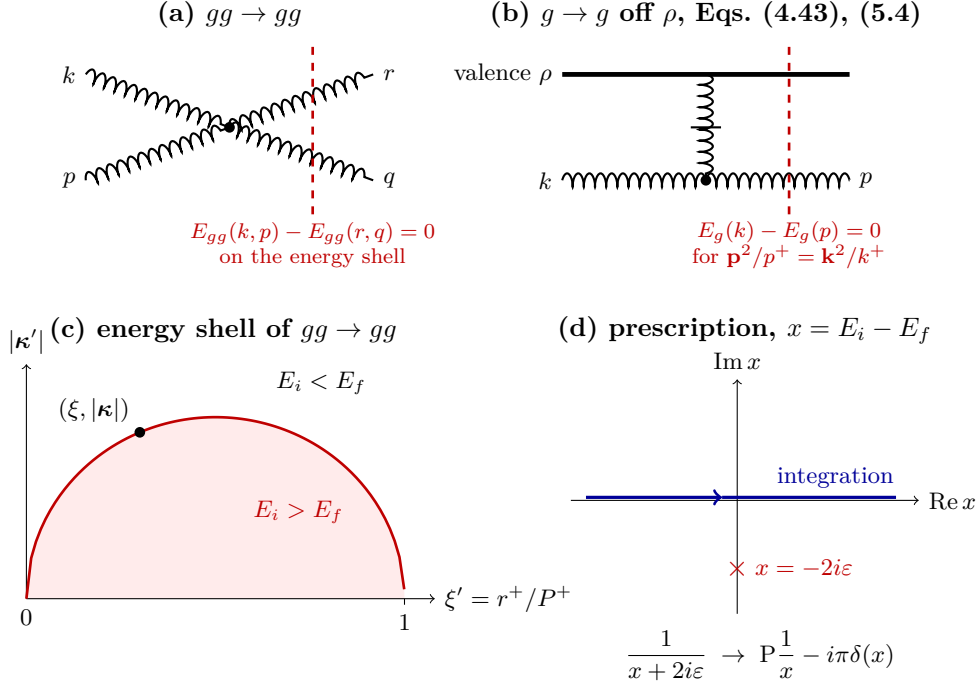


Figure 2: The vanishing energy denominator of Comment 2. (a) Any  $gg \rightarrow gg$  amplitude, whether contact, instantaneous,  $s$ - or  $t$ -channel, carries the overall denominator  $E_{gg}(k, p) - E_{gg}(r, q)$  of the final state (red dashed cut). The  $\delta$ -function fixes  $q = k + p - r$  but not the energy, and the denominator vanishes e.g. at  $r = p$ ,  $q = k$ . (b) The same happens for gluon-number-preserving scattering off the valence background, e.g. the instantaneous term (4.43) and  $\mathbb{B}_3$  in (5.4). (c) With  $\xi = k^+/P^+$  and  $\kappa = (1 - \xi)\mathbf{k} - \xi\mathbf{p}$  fixed, the denominator vanishes on the curve  $|\kappa'|^2 = |\kappa|^2 \xi'(1 - \xi')/[\xi(1 - \xi)]$  (red), which lies *inside* the region of integration over the outgoing momenta; it separates configurations with  $E_i > E_f$  (shaded) from those with  $E_i < E_f$ . (d) The adiabatic switching of Eq. (3.7) moves the pole to  $x = -2i\varepsilon$  below the real axis at second order. As  $\varepsilon \rightarrow 0$  this gives a principal value, which is integrable, plus an on-shell  $\delta$ -function term.

| Transition                                          | Equations                              | Coeff.                                       | Pole in the coefficient?                                                                |
|-----------------------------------------------------|----------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------|
| $1 \rightarrow 1$ off the background, connected     | (4.43), (4.44), (4.46), (4.48), (4.55) | $\mathbb{B}_3$                               | yes                                                                                     |
| $2 \rightarrow 2$ , $\rho$ -independent             | (4.82), (4.84), $t/u$ -channel         | $\mathbb{D}_2$                               | yes                                                                                     |
| $1 \rightarrow 1$ , symmetric part $\{\rho, \rho\}$ | (4.50)+(4.52) $\rightarrow$ (4.54)     | $\mathbb{B}_3$                               | no (cancels between orderings)                                                          |
| $2 \rightarrow 2$ , $\rho$ -dependent               | (4.86)–(4.92)                          | $\mathbb{D}_2$                               | no (cancels; products $\mathbb{A}\mathbb{C}$ , $\mathbb{C}^\dagger\mathbb{A}$ in (5.7)) |
| $3 \rightarrow 3$                                   | (4.97), (4.99)                         | $\mathbb{E}_2$                               | no (cancels; $\mathbb{E}_2 \propto \mathbb{C}\mathbb{C}^\dagger$ )                      |
| $2 \rightarrow 4$                                   | (4.94)                                 | $\mathbb{E}_1$                               | no (cancels; $\mathbb{E}_1 \propto \mathbb{C}\mathbb{C}$ )                              |
| vacuum sector                                       | Sec. 4.1                               | $\mathbb{N}$ , $\mathbb{A}$ , $\mathbb{B}_2$ | no ( $E_i = 0$ )                                                                        |
| splittings                                          | (4.57), (4.59), (4.61), (4.63), (4.65) | $\mathbb{C}$ , $\mathbb{D}_1$                | no ( $E_f > E_i$ )                                                                      |

Intermediate-state denominators, e.g.  $E_g(k) - E_{gg}(r, r')$  in the  $t$ -channel or  $E_{gg}(k, p) - E_g(r')$  in Eq. (4.84), involve a single splitting or merging and vanish at most at collinear endpoints, where they are integrable. On the energy shell, the residue of the remaining pole is the on-shell (Born)

scattering amplitude,

$$T_{fi} = V_{fi}^{(2)} + \sum_m \frac{V_{fm} V_{mi}}{E_i - E_m},$$

where  $V^{(2)}$  is the  $\mathcal{O}(g^2)$  part of the interaction ( $H_{ggg}$  and the instantaneous terms) and  $V$  the  $\mathcal{O}(g)$  part. For  $\mathbb{D}_2$  this is the tree-level  $gg \rightarrow gg$  amplitude, and for  $\mathbb{B}_3$  the amplitude for a soft gluon to scatter off the colour field of the valence charges. These amplitudes are non-zero, so the pole is genuine and cannot cancel.

**Version A — with the prescription.** The revised manuscript specifies how the pole is treated. The following paragraph is added to Sec. 3.2:

#### Proposed changes to the manuscript

*“We use non-degenerate (Rayleigh–Schrödinger) perturbation theory. For the continuum of free soft-gluon states, the degenerate configurations  $E_i = E_f$  form a set of measure zero, and the perturbative series is defined in the standard way by the prescription  $E_i - E_f \rightarrow E_i - E_f + i\varepsilon$  at  $n$ -th order,  $\varepsilon \rightarrow 0^+$ . This prescription follows from the adiabatic definition of  $\Omega$ , Eq. (3.7): with it,  $\Omega$  is the Møller operator of scattering theory and  $\Omega|n\rangle$  are the corresponding in-states. The unitarity relations (4.4)–(4.5) then hold identically. A pole at  $E_i = E_f$  survives only in  $\mathbb{B}_3$  and in the  $\rho$ -independent part of  $\mathbb{D}_2$ , where its residue is the on-shell scattering amplitude; all other coefficients are free of it. In the transformed Hamiltonian  $\Omega^\dagger H \Omega$ , the off-diagonal elements between states of different free energy cancel as before. What remains is multiplied by a factor  $i\varepsilon/(E_i - E_f + i\varepsilon)$ , which equals 1 only on the energy shell and vanishes as  $\varepsilon \rightarrow 0$  everywhere else. Since the energy shell is a set of measure zero, this remainder does not contribute to matrix elements between normalisable states, and the diagonalisation of Sec. 6 holds as stated.”*

Explicitly, with the interaction switched on adiabatically,  $H_I(x^+) e^{\varepsilon x^+}$ , the first- and second-order amplitudes are

$$\omega_1(f \leftarrow i) = \frac{V_{fi}}{E_i - E_f + i\varepsilon}, \quad \omega_2(f \leftarrow i) = \sum_m \frac{V_{fm} V_{mi}}{(E_i - E_f + 2i\varepsilon)(E_i - E_m + i\varepsilon)}, \quad (7)$$

and a short calculation, analogous to Eq. (6), gives for every  $\varepsilon > 0$

$$\begin{aligned} \omega_1(i \leftarrow f)^\dagger &= -\omega_1(f \leftarrow i), \\ \omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger &= -\sum_m \frac{V_{fm} V_{mi}}{(E_i - E_m + i\varepsilon)(E_f - E_m - i\varepsilon)} = -\sum_m \omega_1(m \leftarrow f)^\dagger \omega_1(m \leftarrow i). \end{aligned}$$

These are the unitarity conditions (4.4), (4.5); the  $2i\varepsilon$  at second order is what makes the second identity exact. As  $\varepsilon \rightarrow 0$ ,  $1/(x + 2i\varepsilon) \rightarrow \text{P}(1/x) - i\pi\delta(x)$  with  $x = E_i - E_f$  (Fig. 2(d)). The principal value is integrable, since the pole is simple. The  $\delta$ -function term is an on-shell, anti-Hermitian contribution, which does not enter the unitarity relations. The adiabatic construction of unitary light-cone perturbation theory was discussed recently in Ref. [41], which I now cite here.

*Consequence for Sec. 6.* At  $\mathcal{O}(g)$  nothing changes:  $\mathbb{A}$  and  $\mathbb{C}$  have no vanishing denominators, and Eq. (6.5) holds as stated. At  $\mathcal{O}(g^2)$ , with the prescription, the off-diagonal elements of  $\Omega^\dagger H \Omega$  between states of different free energy cancel as before. What remains is multiplied by a factor of the form  $i\varepsilon/(E_i - E_f + i\varepsilon)$ . This factor equals 1 on the energy shell, where the surviving element is the on-shell amplitude  $T_{fi}$  (the coupling between degenerate states that no unitary transformation could remove), and it vanishes as  $\varepsilon \rightarrow 0$  for every other pair of states. Since the energy shell is a set of measure zero, this remainder gives no contribution to matrix elements between normalisable states (wave packets). The diagonalisation of Sec. 6, Eq. (6.10), therefore holds as stated. No coefficient computed in the paper changes, apart from the specification of the prescription.

### Proposed changes to the manuscript

1. Sec. 3.2, after Eq. (3.5): add the paragraph quoted above.
2. Sec. 3.3: write Eq. (3.7) in the interaction picture, with light-cone time  $x^+$  and the switching factor  $e^{\varepsilon x^+}$ , and give Eq. (7) with the unitarity identities.
3. After Eqs. (4.43) and (4.82): one sentence noting that the overall denominator vanishes for degenerate configurations (e.g.  $r = p$ ,  $q = k$  in (4.82)) and is understood with the prescription of Sec. 3.2.
4. Sec. 5: state that  $\mathbb{B}_3$  and the  $\rho$ -independent part of  $\mathbb{D}_2$  carry this prescription, and that in all other coefficients the overall denominator either never vanishes or cancels between time orderings.
5. Sec. 6: one sentence stating that the off-diagonal elements between degenerate states do not contribute as  $\varepsilon \rightarrow 0$ , so that Eq. (6.10) holds.

**Version B — without introducing a prescription.** We have checked that none of the results derived in the paper depends on how these zeros are treated, and the revised manuscript explains why:

- (i) *Unitarity.* For  $f \neq i$ , the Hermitian part of the  $\mathcal{O}(g^2)$  coefficients is pole-free:

$$\frac{1}{2}[\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger] = -\frac{1}{2} \sum_m \omega_1(m \leftarrow f)^\dagger \omega_1(m \leftarrow i).$$

The pole sits entirely in the anti-Hermitian part,

$$\frac{1}{2}[\omega_2(f \leftarrow i) - \omega_2(i \leftarrow f)^\dagger] = \frac{1}{E_i - E_f} \left[ V_{fi}^{(2)} + \frac{1}{2} \sum_m V_{fm} V_{mi} \left( \frac{1}{E_i - E_m} + \frac{1}{E_f - E_m} \right) \right],$$

i.e. in the generator  $\mathbb{G}$  of Eq. (5.11). The normalisation conditions and the unitarity relations (4.9)–(4.14) involve only the Hermitian parts.

- (ii) *Diagonalisation.* In Sec. 6 the  $\mathcal{O}(g^2)$  coefficients enter  $\Omega^\dagger H \Omega$  only through  $H_0 \Omega^{(2)} + \Omega^{(2)\dagger} H_0$ , whose matrix elements are

$$E_f \omega_2(f \leftarrow i) + E_i \omega_2(i \leftarrow f)^\dagger = -(E_i - E_f) \omega_2(f \leftarrow i) + E_i [\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger].$$

The pole is multiplied by its own zero, and the remainder is the Hermitian part. The cancellation of the off-diagonal elements leading to Eq. (6.10) therefore holds for all pairs of states with  $E_i \neq E_f$  without any regularisation; the exactly degenerate pairs form a set of measure zero.

- (iii) *The coefficients themselves.*  $\mathbb{B}_3$  and  $\mathbb{D}_2$  are therefore given as functions of the momenta for  $E_i \neq E_f$ . Their extension to the energy shell, where the residue is the on-shell amplitude  $T_{fi}$ , requires a boundary condition (a choice of  $i\varepsilon$ ) only when they are integrated across the energy shell in an observable that does not commute with  $H_0$ . This first happens in the computation of single-inclusive gluon production at NLO [44], where the appropriate boundary condition will be specified.

The following paragraph, summarising (i)–(iii), is added after Eq. (4.82):

### Proposed changes to the manuscript

*“The overall energy denominator of the coefficients  $\mathbb{B}_3$  and  $\mathbb{D}_2$  vanishes for degenerate configurations, e.g.  $r = p$ ,  $q = k$  in Eq. (4.82). These configurations form a set of measure*

zero, and none of the results derived here depends on how they are treated. By unitarity, the Hermitian parts of these coefficients are free of the pole,  $\Omega^{(2)} + \Omega^{(2)\dagger} = -\Omega^{(1)\dagger}\Omega^{(1)}$ , and only the Hermitian parts enter the normalisation conditions and the unitarity relations (4.9)–(4.14). In the diagonalisation of Sec. 6 the coefficients enter only in the combination  $(E_i - E_f) \times \text{coefficient}$ , in which the pole cancels identically. The coefficients are therefore given here as functions of the momenta for  $E_i \neq E_f$ . On the energy shell their anti-Hermitian part has a simple pole, whose residue is the on-shell scattering amplitude. A boundary condition (a choice of  $i\epsilon$ ) has to be specified when these coefficients are integrated over the energy shell in an observable, as in the computation of single-inclusive production at NLO [44], where it will be given.”

### Proposed changes to the manuscript

1. After Eq. (4.82): add the paragraph quoted above. After Eq. (4.43): one sentence referring to it.
2. Sec. 5: state that  $\mathbb{B}_3$  and  $\mathbb{D}_2$  are given for  $E_i \neq E_f$  and that their Hermitian parts are pole-free; in all other coefficients the overall denominator either never vanishes or cancels between time orderings.
3. Sec. 6: one sentence showing the combination  $-(E_i - E_f)\omega_2 + E_i(\omega_2 + \omega_2^\dagger)$ , so that the cancellation of the off-diagonal elements visibly requires no regularisation.

### Note to the author (not for the referee)

*Which version to send.* Both are mathematically correct (Part III.2.5 and III.2.6 derive every statement), but they answer different questions. Version A answers the referee’s request (“how this is done should be discussed”) directly. It costs one paragraph and no recalculation, and with it Sec. 6 holds as stated. Version B shows that nothing *in this paper* depends on the treatment, but it postpones the actual regularisation to [44]. Since the referee asked for it explicitly, a second round is likely. My recommendation is Version A. Optionally, add point (ii) of Version B as one extra sentence, because it shows that the results do not depend on the choice of prescription.

*Two corrections to the previous draft.* (1) The pole survives only in  $\mathbb{B}_3$  and in the  $\rho$ -independent part of  $\mathbb{D}_2$ .  $\mathbb{E}_2$  and the  $\rho$ -dependent part of  $\mathbb{D}_2$  are free of it, because their two time orderings combine into a product. The earlier draft listed  $\mathbb{E}_2$  as affected. (2) With the prescription, the couplings between degenerate states do not contribute to matrix elements between normalisable states, so Sec. 6 holds as stated. The earlier phrase “ $\Omega^\dagger H \Omega$  commutes with  $H_0$ ” was correct but weaker than necessary, and has been replaced.

## Comment 3: running coupling and the $\beta$ -function coefficient in (4.78)

### Referee, comment 3

At this order in perturbation theory the QCD coupling should run and the NLO result should depend on a scale associated with the coupling constant renormalization. Indeed one has here a  $\beta$ -function coefficient in (4.78), but somehow no running of the coupling constant. This should be addressed in the text.

**Reply.** I thank the referee for this remark. First, the result itself does not change. The transverse integral  $\int d^2\tilde{p}/\tilde{p}^2$  in Eq. (4.79) is scaleless, so in dimensional regularisation it vanishes,

$$\mu^{2\epsilon} \int \frac{d^{2-2\epsilon}\tilde{p}}{\tilde{p}^2} = 0.$$

The purely gluonic overlap Eq. (4.79) therefore does not contribute to the normalisation coefficient, and Eq. (4.80) is correct as it stands.

The referee’s point concerns what this zero means, and I agree it must be stated in the text. The integral vanishes because its UV and IR divergences cancel, not because it is finite:

$$0 = \mu^{2\epsilon} \int \frac{d^{2-2\epsilon}\tilde{p}}{\tilde{p}^2} = \pi \left( \frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}} \right) \Big|_{\epsilon_{\text{UV}}=\epsilon_{\text{IR}}}. \quad (8)$$

The prefactor that multiplies this integral in Eq. (4.79) comes from Eq. (4.78) and contains the constant  $11/6$ . Its UV part is therefore

$$\frac{g^2 N_c}{8\pi^3} \left( -\frac{11}{6} \right) \frac{\pi}{\epsilon_{\text{UV}}} = -\frac{g^2}{16\pi^2} \frac{\beta_0}{\epsilon_{\text{UV}}}, \quad \beta_0 = \frac{11}{3} N_c \quad (N_f = 0).$$

This is the one-loop gluon field-strength renormalisation. In light-cone (axial-type) gauge it also fixes the renormalisation of the coupling ( $Z_g = Z_3^{-1/2}$ ), which is why the  $\beta$ -function coefficient appears in Eq. (4.78). The additional  $2 \log(k^+/\Lambda)$  term is the light-cone-gauge rapidity (small- $x$ ) divergence. The  $1/\epsilon_{\text{IR}}$  pole has collinear origin.

So the vanishing of Eq. (4.79) is the familiar statement that on-shell massless self-energies vanish in dimensional regularisation. The UV divergence has not disappeared: it is absorbed into the renormalised coupling  $g(\mu)$  ( $\overline{\text{MS}}$ ). The IR (collinear) divergence cancels against real emission in infrared-safe quantities. The renormalisation scale  $\mu$  therefore does not show up in  $\mathbb{N}$  at  $\mathcal{O}(g^2)$ ; it enters through the coupling. The  $\mathcal{O}(g)$  coefficients  $\mathbb{A}$  and  $\mathbb{C}$  are tree level and are to be read as functions of  $g(\mu)$ . Their one-loop,  $\mathcal{O}(g^3)$ , corrections (Ref. [17]) carry the compensating  $\log \mu^2$  with coefficient  $\propto \beta_0$ . Together these turn the fixed coupling of the Weizsäcker–Williams amplitude into a running coupling evaluated at a scale set by the gluon transverse momentum, as in the running-coupling BK/JIMWLK kernels (Balitsky; Kovchegov and Weigert). The running thus becomes explicit in the physical observable, single inclusive gluon production at NLO [44], which combines the present  $\mathcal{O}(g^2)$  coefficients with the  $\mathcal{O}(g^3)$  ones. The same remarks apply to Eq. (6.8), which is the one-loop gluon self-energy and likewise vanishes as a scaleless integral.

#### Proposed changes to the manuscript

1. Keep Eq. (4.80) unchanged. After Eq. (4.79), add a paragraph stating that the scaleless integral vanishes because of the UV/IR cancellation (8). Identify the  $11/6$  in Eq. (4.78) with  $\beta_0/(2N_c)$ , and state that the UV part is the coupling (field-strength) renormalisation, absorbed into  $g(\mu)$  in  $\overline{\text{MS}}$ .
2. Below Eq. (5.2), state that  $g \equiv g(\mu)$ , and that the running coupling appears in physical observables once the  $\mathcal{O}(g^3)$  coefficients of Ref. [17] are included. Add a sentence to the conclusions.
3. After Eq. (6.8), add the same remark (scaleless one-loop self-energy).
4. Add references: I. Balitsky, Phys. Rev. D 75 (2007) 014001 [hep-ph/0609105]; Yu. V. Kovchegov and H. Weigert, Nucl. Phys. A 784 (2007) 188 [hep-ph/0609090]; G. Leibbrandt, Rev. Mod. Phys. 59 (1987) 1067 (non-covariant gauges).

#### Note to the author (not for the referee)

Your treatment is kept: the integral is zero and  $\mathbb{N}$  in Eq. (4.80) does not change. Equation (8) is not a new, non-zero contribution. It only splits that zero into its UV and IR parts, to show the referee where the  $\beta$ -function coefficient and the coupling renormalisation sit. If you prefer not to write the  $1/\epsilon$  poles at all, a sentence of the form “this zero reflects the cancellation of a UV divergence, which renormalises the coupling (coefficient  $\beta_0$ ), against a collinear IR

divergence” is enough to answer the referee. Please verify the bibliographic data of the added references.

#### Comment 4: “operator-valued coefficients” in Sec. 3.1

##### Referee, comment 4

[...] In Sec 3.1 it is stated that the “coefficients are not ordinary c-number functions but operator quantities acting in the valence Hilbert space”. [...] The full Hilbert space of the theory is of course a complex-coefficient vector space [...]. The statement about noncommuting operator coefficients only applies to the soft gluon part of the Hilbert space, not the full one. The fact that the states in e.g. eq (3.2) are states in only the soft subspace should be emphasized more in the text.

**Reply.** I agree that the wording was misleading. The full Hilbert space is  $\mathcal{H} = \mathcal{H}_{\text{soft}} \otimes \mathcal{H}_{\text{val}}$ , an ordinary complex vector space. In any complete basis  $|n\rangle_{\text{soft}} \otimes |v\rangle_{\text{val}}$  all expansion coefficients are c-numbers. The coefficients in Eq. (3.2) and Eq. (4.1) are operator-valued only because the expansion is performed in the Fock basis of the *soft* subspace alone, with the valence degrees of freedom left unexpanded; the coefficients then act on  $\mathcal{H}_{\text{val}}$  through  $\rho$ . The paragraph is rewritten as follows (new text):

##### Proposed changes to the manuscript

*“The expansion (3.2) is an expansion in the Fock basis of the soft subspace  $\mathcal{H}_{\text{soft}}$  only. The full Hilbert space  $\mathcal{H}_{\text{soft}} \otimes \mathcal{H}_{\text{val}}$  is an ordinary complex vector space, and in a complete basis  $|n\rangle \otimes |v\rangle$  all coefficients are c-numbers. We choose not to expand the valence part, which is the natural choice in the Born–Oppenheimer treatment. The coefficients of the soft Fock states are then functionals of the valence charge density  $\rho$ , i.e. operators on  $\mathcal{H}_{\text{val}}$ . Since the  $\rho$ ’s obey the algebra (2.19), these operator-valued coefficients do not commute in general, and their ordering must be kept. All states written in the form (3.2), and the vacuum  $|0\rangle \equiv |0\rangle_{\text{soft}} \otimes |v\rangle$  (see App. B), should be understood in this sense.”*

In addition, the definition  $|0\rangle \equiv |0\rangle \otimes |v\rangle$  now given only in App. B is moved into Sec. 3.1. The first sentence of Sec. 4 (“The perturbative coefficients are not ordinary c-number functions...”) is rephrased in the same way.

## Part II — Further errors found in the manuscript

The following items were found in a complete re-reading. Group A contains substantive points; the first four affect statements in the paper, not only typography. Group B lists equation-level errors. Group C lists presentation and reference issues. A short version of B and C (“Other changes”) can be appended to the reply.

### A. Substantive points

#### A1. Normalisation of the one-gluon state, Eqs. (4.66)–(4.76).

- (a) The component  $|(\Psi_i^a(k))_{gg\rho}\rangle$  appears in Eqs. (4.66), (4.68), (4.70), (4.75), but it is *not defined* in Sec. 4.2. From its use in Eq. (4.75) it must be the disconnected  $\mathcal{O}(g)$  emission by the valence, with the incoming gluon as a spectator (the term  $\mathbb{A} a^\dagger$ , which is omitted from Eq. (4.16)). It should be defined explicitly.

- (b) The statement that  $\langle (\Psi_i^{a'}(k'))_{gg\rho} | (\Psi_i^a(k))_{gg} \rangle + \text{h.c.} = 0$  “identically” (Eq. (4.68)) cannot hold as a mutual cancellation. The first term requires  $k^+ > k'^+$  (splitting  $k \rightarrow k' + (k - k')$  with  $(k - k')^+ > 0$ ), while its partner requires  $k'^+ > k^+$ . The two terms have disjoint support, so each is separately non-zero. [*Derived from plus-momentum positivity.*]
- (c) Condition Eq. (4.67) with  $k \neq k'$  also receives  $\rho$ -linear contributions from the overlap of the bare state  $|g(k')\rangle$  with the  $\mathcal{O}(g^2)$  one-gluon components Eqs. (4.43), (4.44), (4.46), (4.48) and Eq. (4.55). These are not included in Eq. (4.70). The off-diagonal ( $k \neq k'$ ) part of Eq. (4.67) is precisely the unitarity relation Eq. (4.10) for  $\mathbb{B}_3$ , and should be shown to hold (item (b) terms cancelling against these). Only the  $\delta^3(k - k')$  part determines  $\mathbb{N}$ .

**A2. Normalisation of the two- and three-gluon states (Secs. 4.3.3, 4.4.1).** The statement “Since the interacting part of the wave function itself starts at order  $g^2$ , corrections to the normalization factor ... contribute only at higher order” is incorrect as written. Eq. (4.17) shows that  $\Omega|gg\rangle$  has  $\mathcal{O}(g)$  components ( $\mathbb{A}^\dagger$  absorption,  $\mathbb{A}$  emission,  $\mathbb{C}$  splitting,  $\mathbb{C}^\dagger$  merging), whose squares contribute at  $\mathcal{O}(g^2)$  to the norm. It is only the *new, connected*  $\mathcal{O}(g^2)$  components computed in Secs. 4.3–4.4 that do not affect the norm at this order. The same  $\mathbb{N}$  multiplies all sectors because  $\Omega$  is a single operator, and its value is fixed by unitarity. The  $\mathcal{O}(g)$  components should also be listed (or referred to) in Eqs. (4.95) and (4.100).

**A3. Section 6 (diagonalisation).**

- (a) Eq. (6.4): the sign of the first term is wrong. With  $\Omega^{(1)} \ni \mathbb{A} a^\dagger - \mathbb{A}^\dagger a$  and  $[H_0, a^\dagger(p)] = \frac{\mathbf{p}^2}{2p^+} a^\dagger(p)$  one gets  $+\mathbb{A} \frac{\mathbf{p}^2}{2p^+} a^\dagger + \mathbb{A}^\dagger \frac{\mathbf{p}^2}{2p^+} a$ . Only with the  $+$  sign does the term cancel  $H_g$ , using Eq. (5.2):  $\mathbb{A} \mathbf{p}^2/2p^+ = -g \mathbf{p}^j \rho(-\mathbf{p})/(\sqrt{2} p^{+3/2})$ . The sign convention of the  $\mathbb{C}$  terms in the same equation is also consistent only with  $+$ . [*Derived.*]
- (b) Eq. (6.5) holds only through  $\mathcal{O}(g)$ . The sentence after it (“the classical WW energy ... is already described in the choice of  $\mathbb{A}$ ”) is confusing, because the  $\rho^2$  energy is  $\mathcal{O}(g^2)$  and appears only in Eq. (6.10).
- (c) Eq. (6.7) keeps only the vacuum-energy term and one  $\mathbb{C}^\dagger \mathbb{C}$  term. At  $\mathcal{O}(g^2)$ ,  $\Omega^\dagger H \Omega$  also receives  $\rho$ -linear and  $\rho\rho$  terms of the form  $a^\dagger a$ , from  $[H_g, \mathbb{C}]$ ,  $[H_{ggg}, \mathbb{A}]$ ,  $H_{gg\text{-inst}}$  and  $[H_g, \mathbb{A}]$ , and  $a^\dagger a^\dagger a a$  terms, from  $H_{gggg}$ , the instantaneous term and  $[H_{ggg}, \mathbb{C}]$  including the  $t$ -channel. Their off-diagonal parts cancel against  $[H_0, g^2 \Omega^{(2)}]$  for all  $E_i \neq E_f$ , and the degenerate remainder is handled as in the reply to Comment 2. The text should show these cancellations; at present (6.7) reads as if these terms were absent.
- (d) Eq. (6.8) is the one-loop gluon self-energy. It vanishes only as a scaleless integral in dimensional regularisation, and it carries the same UV pole as Eq. (4.79) (Comment 3).

**A4. Definition of  $\Omega$ , Eq. (3.7).** As written,  $T \exp(-i \int_{-\infty}^0 dt H(t))$  uses the full Schrödinger-picture Hamiltonian and ordinary time  $t$ . For the adiabatic theorem/Gell-Mann–Low construction one needs the interaction-picture Hamiltonian with light-cone time and a switching factor,  $U_\varepsilon(0, -\infty) = T \exp(-i \int_{-\infty}^0 dx^+ e^{\varepsilon x^+} H_I(x^+))$ , and a normalisation that removes the divergent phase. The notation  $U(-\infty, 0)$  vs.  $U(0, -\infty)$  should also be fixed. This is directly relevant to Comment 2.

**A5. Plus momentum carried by  $\rho$ .** In definition Eq. (2.18) both valence operators carry the same  $k^+$ , so  $\rho$  transfers only transverse momentum. Yet Fig. 1 and Eqs. (4.42)–(4.48) assume that the valence exchanges plus momentum with the soft gluon, restricted to  $\Lambda < |p^+ - k^+| < \vee$ . The paper should explain how the plus-momentum transfer arises from the eikonal Hamiltonian. It should also note that the window  $|p^+ - k^+| > \Lambda$  is what regularises the  $1/(p^+ - k^+)^2$  singularity of the instantaneous vertex Eq. (D.5), so the results of Eqs. (4.43), (4.44) and (5.4) depend on it. (Fig. 1’s caption mentions only  $\Lambda < p^+ - k^+ < \vee$ ; Eq. (4.44) uses  $\Lambda < k^+ - p^+.$ )

**A6. Power counting.** The expansion  $\Omega = 1 + g\Omega^{(1)} + g^2\Omega^{(2)}$  counts  $\rho = \mathcal{O}(1)$ , i.e. the dilute regime. In the saturation regime  $\rho \sim 1/g$  and  $g\rho = \mathcal{O}(1)$ , and the classification into “ $\mathcal{O}(g)$ ” and “ $\mathcal{O}(g^2)$ ” changes (e.g.  $g^2\rho\rho$  terms are  $\mathcal{O}(1)$ ). One or two sentences stating the counting would prevent confusion, e.g. with the phrase “at leading order in the coupling and the highest order of  $\rho$ ” in Sec. 3.3. The notation should also be made consistent: in Eq. (4.2) the  $g$  is factored out, while the footnote states that  $\mathbb{A}$  itself is  $\mathcal{O}(g)$ .

**A7. Appendix E.** (a) The denominators in Eq. (E.3) contain  $E_i^{(1)}$  and  $E_i^{(2)}$ , which is Brillouin–Wigner-like and not the Rayleigh–Schrödinger theory quoted. (b) The symbol  $|n^{(i)}\rangle$  is used both for the  $i$ -th order correction and for intermediate basis states. (c) The degenerate case (Comment 2) is not addressed. (d) Despite its title, the appendix does not show where operator ordering matters. For example,  $E_i^{(1)}$  is itself an operator on  $\mathcal{H}_{\text{val}}$ , and Eq. (4.28) is the place where ordering enters. A rewrite along the lines of Eq. (6)/Eq. (7) would serve better.

## B. Equation-level errors

| Eq.            | As printed                                                                       | Correction                                                                                                                                                                                                                                                                                                             | Status           |
|----------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| (4.6)          | $\mathbb{N}_2 = 1 - \int \mathbb{A}^\dagger \mathbb{A}$                          | $\mathbb{N}_2 = -\frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \mathbb{A}^\dagger \mathbb{A}$ (consistent with the second equality and with Eq. (5.1))                                                                                                                                                                       | Derived          |
| (4.17), line 1 | $-\frac{1}{(2\pi)^{3/2}} \mathbb{A}_j^{\dagger b}(p)  g_k^c(q)\rangle$           | $\dots  g_i^a(k)\rangle$ (absorbing $p$ leaves $k$ )                                                                                                                                                                                                                                                                   | Derived          |
| (4.17), line 2 | $\int \frac{d^3 q}{(2\pi)^3} \mathbb{A}_k^c(q)  ggg\rangle$                      | missing factor $(2\pi)^{3/2}$ , since $a^\dagger  gg\rangle = (2\pi)^{3/2}  ggg\rangle$ by (B.4), (B.5); cf. Eq. (4.15)                                                                                                                                                                                                | Derived          |
| (4.31), (4.36) | $-ig^2 f^{abc} \dots$ (commutator term)                                          | Inserting Eq. (4.29) into Eq. (4.27) gives the same expression with $+ig^2 f^{abc}$ (from $[\rho^b(-\mathbf{k} + \mathbf{p}), \rho^c(-\mathbf{p})] = if^{bcd} \rho^d(-\mathbf{k})$ ); the sign should be checked                                                                                                       | To verify (sign) |
| (4.36)         | sum of (4.31) and (4.22) only                                                    | the instantaneous term Eq. (4.24), listed in Eq. (4.35), is missing                                                                                                                                                                                                                                                    | Derived          |
| (4.55)         | $-\frac{g^2 f^{abc}}{8\pi^3} \int dp^+ \dots$                                    | $+\frac{ig^2 f^{abc}}{8\pi^3} \int \frac{dp^+}{\sqrt{k^+ p^+}} \dots$ : the factor $1/\sqrt{k^+ p^+}$ is required by dimensions (compare Eq. (4.54)), and the $i$ comes from $[\rho^b(-\mathbf{p}), \rho^a(\mathbf{k})] = -if^{abc} \rho^c(\mathbf{k} - \mathbf{p})$ . Same correction in the second line of Eq. (5.4) | Derived          |
| (4.61)         | $(\frac{v^+ + p^+}{q^+} \mathbf{k}^k - \mathbf{p}^k - \mathbf{v}^k) \delta_{jn}$ | $\mathbf{k}^k \rightarrow \mathbf{q}^k$ (vertex for $v \rightarrow p + q$ , cf. Eq. (D.3))                                                                                                                                                                                                                             | Derived          |
| (4.60)         | factor “ $2 \times \frac{1}{12}$ ”                                               | check against (4.62): the matrix element $\langle ppp   H_{ggg}   uv \rangle$ already contains the splitting of either gluon                                                                                                                                                                                           | To verify        |
| (4.86)         | $f^{dbe}, \rho^e(-\mathbf{s})$                                                   | $f^{dbc}$ and $\rho^e(\mathbf{s})$ (absorption of $s$ ; cf. Eq. (4.41) and Eq. (4.88))                                                                                                                                                                                                                                 | Derived          |
| (4.94)         | $-\frac{g^4 f^{ade} f^{bfh}}{128\pi^3}$ ; LHS $ (\Psi)_{gg}\rangle$              | $g^4 \rightarrow g^2$ (two $H_{ggg}$ vertices); LHS should be $ (\Psi_{ji}^{ba}(k, p))_{gg}^3\rangle$                                                                                                                                                                                                                  | Derived          |

| Eq.                    | As printed                                                                                                                                                                 | Correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Status                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| (4.97)<br>vs<br>(4.99) | measure $d^3p$ vs $\frac{d^3p}{(2\pi)^3}$                                                                                                                                  | use the same measure in both                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Derived                    |
| (5.3)                  | $(\mathbf{p}^i + \mathbf{k}^i)$                                                                                                                                            | $(\mathbf{p}^i + \mathbf{q}^i)$ ; $k$ is not defined here,<br>$k = p + q$ (cf. Eq. (4.22))                                                                                                                                                                                                                                                                                                                                                                                                                         | Derived                    |
| (5.6)                  | $\frac{\mathbf{q}^2}{2p^+}$ in<br>$(\frac{\mathbf{p}^2}{2p^+} + \frac{\mathbf{q}^2}{2p^+} + \frac{\mathbf{r}^2}{2r^+} - \frac{\mathbf{k}^2}{2k^+})$                        | $\frac{\mathbf{q}^2}{2q^+}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived                    |
| (5.7)                  | $(\frac{\mathbf{p}^2}{2p^+} - \frac{\mathbf{q}^2}{2q^+} - \frac{\mathbf{r}^2}{r^+})$                                                                                       | $\frac{\mathbf{r}^2}{2r^+}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Derived                    |
| (5.7)                  | $s$ -channel term<br>$\int \mathbb{C}(r', k, p) \mathbb{C}^\dagger(r', r, q) w_p$ ,<br>with energy weight<br>$w_p = \frac{E_g(r') - E_g(k) - E_g(p)}{E_f - E_i}$           | Converting (4.84) gives<br>$\int \mathbb{C}(r'; q, r) \mathbb{C}^\dagger(r'; k, p) w$ with<br>$w = \frac{E_g(r) + E_g(q) - E_g(r')}{E_i - E_f}$ (Part III,<br>Step 13). Since $w_p - w = 1$ , the<br>printed term equals the correct one<br>plus the pole-free product $\int \mathbb{C}\mathbb{C}^\dagger$ ; $w_p$<br>is the weight of the reversed process<br>$r q \rightarrow k p$ . The $\mathbb{A}\mathbb{C}$ terms of (5.7) use<br>the correct rule. Colour/polarisation<br>indices are also missing in (5.7) | Derived; please<br>check   |
| (5.7)                  | no $t$ - or $u$ -channel term                                                                                                                                              | add $\mathbb{D}_2^{(t)}$ , Eq. (5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comment 1                  |
| (6.4)                  | $-\mathbb{A}_j^b(p) \frac{\mathbf{p}^2}{2p^+} a_j^{\dagger b}(p)$                                                                                                          | $+\mathbb{A}_j^b(p) \frac{\mathbf{p}^2}{2p^+} a_j^{\dagger b}(p)$ (see A3(a))                                                                                                                                                                                                                                                                                                                                                                                                                                      | Derived                    |
| (2.13)<br>vs (C.6)     | $+\frac{g^2}{2} f f \frac{1}{\partial^+}(\cdot) \frac{1}{\partial^+}(\cdot)$ vs<br>$\frac{g^2}{4} f f (\dots - 2 \frac{1}{\partial^+}(\cdot) \frac{1}{\partial^+}(\cdot))$ | opposite signs of the instantaneous<br>four-gluon term; one is a typo (the<br>standard light-cone Hamiltonian has<br>$+\frac{g^2}{2}$ , consistent with (C.5))                                                                                                                                                                                                                                                                                                                                                     | Derived<br>(inconsistency) |
| (A.5)                  | $H_{LCYM} = P^+$                                                                                                                                                           | $P^-$ (with Sec. 2.1, $p^-$ generates $x^+$<br>translations), or state that $P_+ = P^-$                                                                                                                                                                                                                                                                                                                                                                                                                            | Derived                    |
| (A.1)                  | $[D_\mu, D_\nu]$                                                                                                                                                           | $[D_\mu, D_\nu]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Typo                       |
| (B.5)                  | $E_{gg}(k, p, q)$                                                                                                                                                          | $E_{ggg}(k, p, q)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Typo                       |
| (4.33)                 | $ \Psi\rangle$                                                                                                                                                             | $ \Psi\rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Typo                       |
| (4.74),<br>(4.75)      | $\mathbf{k}'^{2'}$                                                                                                                                                         | $\mathbf{k}'^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Typo                       |
| (4.53)–<br>(4.55)      | superscripts 1, 2 on the $\{\rho, \rho\}$ ,<br>$[\rho, \rho]$ components in (4.53)                                                                                         | drop them, or use them in (4.54),<br>(4.55) as well                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Typo                       |
| (2.21),<br>(2.23)      | stray commas “+ h.c.”;<br>unbalanced bracket in (2.23)                                                                                                                     | remove / balance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Typo                       |

## C. Presentation and references

- C1.** Sec. 4.3: the incoming momenta are labelled  $(k, p)$  in (4.81),  $(p, q)$  in (4.83) and  $(r, s)$  in (4.85)–(4.92), but summed as  $(k, p)$  in Eq. (4.95). Use one labelling throughout.
- C2.** The instantaneous four-gluon Hamiltonian is called  $H_{gggg\text{-inst}}$  in (2.17), (2.24), (6.6) and  $H_{gg\text{-}gg\text{-inst}}$  in (4.81), (D.7). Use one name.
- C3.** Sec. 4.2.5, before Eq. (4.74): the sentence “The only remaining contribution requiring special attention is the overlap of the purely two-gluon sector” is placed before the  $\rho$ -overlaps. It belongs before Eq. (4.77).
- C4.** Sec. 5, first paragraph: the matching also uses  $\Omega|ggg\rangle$  (for  $\mathbb{E}_2$ ), not only  $\Omega|0\rangle, \Omega|g\rangle, \Omega|gg\rangle$ .

**C5.** Introduction, last paragraph: “diagonalize the QCD Hamiltonian” should be “pure Yang–Mills”; Sec. 7 is not described.

**C6.** Eq. (5.11): the phrase “resumming repeated gluon emissions . . . into a single unitary operator” overstates what a truncated  $\mathbb{G}$  does. Suggest “ $\mathbb{G}$  is the Hermitian generator of  $\Omega$ , known here through  $\mathcal{O}(g^2)$ ”.

**C7.** References:

- [20] and [32] are the same paper (Altinoluk & Kovner, PRD 83 (2011)); [38] and [41] are the same paper (Munier, arXiv:2510.05256). Merge them and renumber.
- Introduction: “[20–24, 24–33]” cites [24] twice; write “[20–33]”.
- [7], [8]: “the The Color Glass Condensate” → “the Color Glass Condensate”.
- [53] is Peskin & Schroeder (co-author missing). For the treatment of scaleless integrals a more specific reference (e.g. J. C. Collins, *Renormalization*, CUP 1984) is preferable.
- [50]: “Lifshits” → “Lifshitz”.
- Several journal references lack article/page numbers ([10]–[12], [15], [16], [18]–[23], [25], [27], [29], [31], [35], [36]).
- Conclusions: “. . . requires the explicit coefficients of the evolution operator derived here [54]”. [54] is Kovner & Lublinsky (2006); the intended reference is probably [44]. Please check.
- Add the running-coupling and non-covariant-gauge references listed under Comment 3.

With these changes I believe all the referee’s concerns have been addressed. I thank the referee again for comments that have improved the manuscript.

Sincerely,

Ramkumar Radhakrishnan

## Part III — Detailed calculations for the author (not for the referee)

This part contains the complete calculations behind the replies to Comments 1 and 2, written step by step so that every statement in Part I can be checked. It is meant for you, not for the referee. The key algebraic identities below — the vertex in relative momenta, the two-body and splitting energies, the polarisation tensor  $P_{ilnj}$ , the decomposition of  $\omega_2$ , the  $i\varepsilon$  unitarity identities, the remainder (III.35), the time-ordering identity (III.29) and the  $s$ -channel weight difference of Step 13 — were also checked numerically with random momenta, to machine precision.

Notation used throughout:  $E_g(x) \equiv \mathbf{x}^2/(2x^+)$  is the free light-cone energy of a gluon;  $E_i, E_f, E_m$  are the free energies of the initial, final and intermediate Fock states;  $V$  is the  $\mathcal{O}(g)$  part of the interaction ( $H_g, H_{ggg}$ ) and  $V^{(2)}$  its  $\mathcal{O}(g^2)$  part ( $H_{gggg}, H_{gg\text{-inst}}, H_{gggg\text{-inst}}$ );  $V_{fi} \equiv \langle f|V|i \rangle$ .

### III.1 Comment 1: the $t$ -channel gluon exchange

#### III.1.1 The three kinds of $\mathcal{O}(g^2)$ terms, and your question about (4.82)

You are right that the “usual” (non-instantaneous) terms in Eq. (4.82) carry all three colour structures. But they are not gluon exchanges. At  $\mathcal{O}(g^2)$  there are three different kinds of contributions to the transition of two incoming gluons  $k, p$  into two outgoing gluons  $r, q$ , shown in Fig. 3:

- (a) **Contact interaction.** One vertex of  $H_{gggg}$ , coming from the  $F_{ij}F_{ij}$  part of the Hamiltonian, Eq. (A.6):  $\frac{g^2}{4} f^{abc} f^{ade} A_i^b A_j^c A_i^d A_j^e$ . When the four fields are contracted with the four external gluons,  $f^{abc} f^{ade}$  produces the three colour structures  $f^{a'ad} f^{a'cb}$ ,  $f^{a'ac} f^{a'db}$ ,  $f^{a'ab} f^{a'dc}$ , each multiplied by a  $\delta\delta - \delta\delta$  polarisation factor, as in (4.82). This is the light-cone version of the covariant four-gluon vertex,

$$-ig^2 [f^{abe} f^{cde} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}) + f^{ace} f^{bde} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho}) + f^{ade} f^{bce} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma})],$$

which also has three colour structures and is still *one* diagram. No gluon propagates between two vertices. (Check: for  $f^{a'ad} f^{a'cb}$  the pattern  $g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}$  with legs  $(a, i), (d, l), (c, n), (b, j)$  gives  $\delta_{in}\delta_{lj} - \delta_{ij}\delta_{ln}$ , which is the  $\delta_{ik}\delta_{jl} - \delta_{lk}\delta_{ij}$  of (4.82) with the paper’s label  $k$  for the polarisation  $n$  of  $q$ .)

- (b) **Instantaneous exchange.** One vertex of  $H_{gggg\text{-inst}}$ , coming from  $\Pi^2$ , Eq. (A.7):

$$\frac{g^2}{2} f^{abc} f^{ade} \frac{1}{\partial^+} (A_i^b \partial^+ A_i^c) \frac{1}{\partial^+} (A_j^d \partial^+ A_j^e).$$

Each current  $A\partial^+ A$  joins two gluons with the *same* polarisation index, and  $1/(\partial^+)^2$  gives  $1/(\text{exchanged plus momentum})^2$ . This produces one term per channel — the plus-momentum ratios in (4.82), written here with the paper’s labels (the polarisation index of  $q$  is called  $k$ ):

$$\begin{array}{lll} t: f^{a'ad} f^{a'cb} & \frac{(q^+ + p^+)(r^+ + k^+)}{(k^+ - r^+)(p^+ - q^+)} \delta_{il} \delta_{jk} & \text{currents } k \rightarrow r, p \rightarrow q \\ u: f^{a'ac} f^{a'db} & \frac{(p^+ + r^+)(q^+ + k^+)}{(k^+ - q^+)(p^+ - r^+)} \delta_{lj} \delta_{ik} & \text{currents } k \rightarrow q \text{ and } p \rightarrow r \\ s: f^{a'ab} f^{a'dc} & \frac{(q^+ - r^+)(k^+ - p^+)}{(k^+ + p^+)(r^+ + q^+)} \delta_{ij} \delta_{lk} & \text{currents } (k, p) \text{ and } (r, q) \end{array}$$

Since  $p^+ - q^+ = r^+ - k^+$ , the  $t$ -channel denominator is  $-(k^+ - r^+)^2$ : one over the square of the plus momentum carried by the exchanged gluon.

- (c) **Propagating exchange.** Two vertices of  $H_{ggg}$ , each  $\mathcal{O}(g)$ , at *different* light-cone times, with an intermediate state in between. For the  $s$ -channel this is (4.84) (one-gluon intermediate state). For the  $t$ - and  $u$ -channels it is missing.

Only (c) has a gluon that propagates between two vertices; (a) and (b) are single vertices acting at one instant. Eq. (4.82) contains (a) and (b) in all three colour structures, and the referee's “ $t$ -channel gluon exchange ... where one passes through a three-gluon intermediate state” is (c) in the  $t$ - and  $u$ -channels.

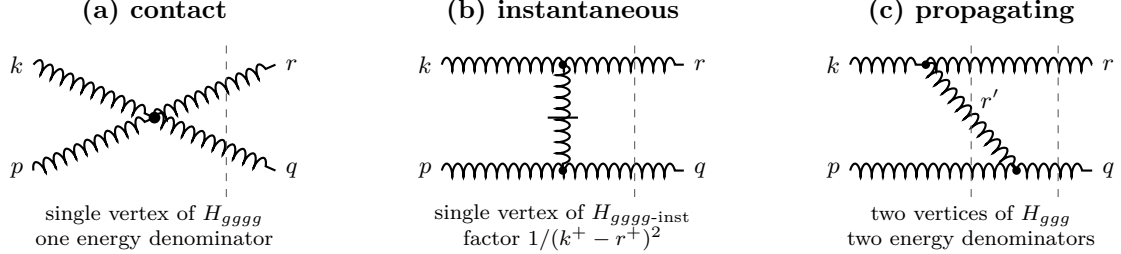


Figure 3: The three kinds of  $\mathcal{O}(g^2)$  contributions to the transition of two incoming soft gluons  $k, p$  into two outgoing gluons  $r, q$  (light-cone time runs from left to right; dashed lines mark intermediate and final states). (a) Contact interaction: a single vertex of  $H_{gggg}$ , from  $F_{ij}F_{ij}$ . It carries the three colour structures of Eq. (4.82) with the  $\delta\delta - \delta\delta$  polarisation factors, but no gluon is exchanged. (b) Instantaneous exchange (crossed line): a single vertex of  $H_{gggg\text{-inst}}$ , from  $\Pi^2$ , with the factor  $1/(\text{exchanged plus momentum})^2$ . Eq. (4.82) contains it in the  $s$ -,  $t$ - and  $u$ -channels. (c) Propagating exchange: two  $H_{ggg}$  vertices at different light-cone times, with an intermediate state and its own energy denominator. The  $s$ -channel version is Eq. (4.84). The  $t$ - and  $u$ -channel versions, shown here for the  $t$ -channel, are missing from Sec. 4.3.1.

### III.1.2 Why a gluon exchange has an instantaneous and a propagating part

In light-cone gauge  $A^+ = 0$  the gluon propagator is

$$D_{ab}^{\mu\nu}(k) = \frac{i\delta_{ab}d^{\mu\nu}(k)}{k^2 + i0}, \quad d^{\mu\nu}(k) = -g^{\mu\nu} + \frac{n^\mu k^\nu + k^\mu n^\nu}{n \cdot k}, \quad (\text{III.1})$$

with  $n \cdot k = k^+$  ( $n$  has only a minus component,  $n^- = 1$ ). Let  $\tilde{k}$  be the on-shell momentum with the same  $k^+$  and  $\mathbf{k}$ , i.e.  $\tilde{k}^- = \mathbf{k}^2/(2k^+)$ . The two physical polarisation vectors  $\epsilon_\lambda$  of the on-shell gluon obey the completeness relation

$$\sum_{\lambda=1,2} \epsilon_\lambda^\mu \epsilon_\lambda^{\nu*} = -g^{\mu\nu} + \frac{n^\mu \tilde{k}^\nu + \tilde{k}^\mu n^\nu}{n \cdot k}. \quad (\text{III.2})$$

Subtracting (III.2) from  $d^{\mu\nu}$  gives  $d^{\mu\nu} - \sum_\lambda \epsilon_\lambda^\mu \epsilon_\lambda^{\nu*} = [n^\mu(k - \tilde{k})^\nu + (k - \tilde{k})^\mu n^\nu]/k^+$ . The vector  $k - \tilde{k}$  has only a minus component,  $k^- - \mathbf{k}^2/(2k^+) = k^2/(2k^+)$  (using  $k^2 = 2k^+k^- - \mathbf{k}^2$ ), so  $(k - \tilde{k})^\mu = \frac{k^2}{2k^+} n^\mu$ . Therefore

$$\frac{d^{\mu\nu}(k)}{k^2 + i0} = \sum_\lambda \frac{\epsilon_\lambda^\mu \epsilon_\lambda^{\nu*}}{k^2 + i0} + \frac{n^\mu n^\nu}{(k^+)^2}. \quad (\text{III.3})$$

The first term has a pole and describes a physical gluon propagating between two vertices. The second has no pole in  $k^-$  and describes an instantaneous interaction. To see this, Fourier transform to light-cone time  $x^+$  at fixed  $k^+ > 0$ :

$$\begin{aligned} \int \frac{dk^-}{2\pi} \frac{i e^{-ik^-x^+}}{2k^+k^- - \mathbf{k}^2 + i0} &= \theta(x^+) \frac{e^{-i\tilde{k}^-x^+}}{2k^+}, \\ \int \frac{dk^-}{2\pi} \frac{i e^{-ik^-x^+}}{(k^+)^2} &= \frac{i\delta(x^+)}{(k^+)^2}. \end{aligned} \quad (\text{III.4})$$

In the first integral the pole  $k^- = \tilde{k}^- - i0$  lies below the real axis. For  $x^+ > 0$  the factor  $e^{-ik^-x^+}$  decays in the lower half plane, so the contour is closed there, picks up the pole, and the residue theorem gives  $(-2\pi i) \cdot \frac{1}{2\pi} \cdot \frac{i e^{-i\tilde{k}^-x^+}}{2k^+} = \frac{e^{-i\tilde{k}^-x^+}}{2k^+}$ . For  $x^+ < 0$  the contour is closed above and the result is zero. (For  $k^+ < 0$  the pole is above the axis, and the gluon propagates backward,  $\theta(-x^+)$ .)

Two conclusions follow:

- A gluon line with positive plus momentum always runs forward in light-cone time. In the  $t$ -channel this decides which vertex comes first: if  $r'^+ = k^+ - r^+ > 0$ , the gluon is emitted by  $k$  and absorbed by  $p$  (Fig. 1(a)); if  $k^+ - r^+ < 0$ , it is emitted by  $p$  and absorbed by  $k$  (Fig. 1(b)). In the  $s$ -channel there is only one ordering: the other one would need  $r, q$  and an extra gluon to be created from nothing, which is impossible with positive plus momenta.
- The  $\delta(x^+)$  in the second term is an interaction at a single instant: this is the origin of the instantaneous terms of  $H_{gggg\text{-inst}}$ , with their  $1/(k^+)^2$ .

So a covariant gluon exchange equals its propagating part plus its instantaneous part. In the paper:

| Covariant diagram     | Light-cone content                                                                                                   | In the paper                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| four-gluon vertex     | contact term of $H_{gggg}$ (from $F_{ij}F_{ij}$ )                                                                    | (4.82), $\delta\delta$ terms                     |
| $s$ -channel exchange | propagating: one ordering, one-gluon intermediate state; plus instantaneous                                          | (4.84); (4.82), $f^{a'ab}f^{a'dc}$ term          |
| $t$ -channel exchange | propagating: two orderings, three-gluon intermediate state, selected by the sign of $k^+ - r^+$ ; plus instantaneous | <b>missing</b> ; (4.82), $f^{a'ad}f^{a'cb}$ term |
| $u$ -channel exchange | as $t$ , with $r \leftrightarrow q$                                                                                  | <b>missing</b> ; (4.82), $f^{a'ac}f^{a'db}$ term |

### III.1.3 Four checks that (4.82) does not contain the propagating exchange

1. *Vertices.*  $H_{gggg}$  and  $H_{gggg\text{-inst}}$  are single  $\mathcal{O}(g^2)$  vertices, and (4.81) contains one matrix element of them: (4.82) is first order. The propagating exchange is second order in  $H_{ggg} = \mathcal{O}(g)$ : it contains two matrix elements.
2. *Energy denominators.* (4.82) has one energy denominator,  $1/[E_{gg}(k, p) - E_{gg}(r, q)]$ . The propagating exchange has two: the same overall one and the intermediate one,  $1/[E_g(k) - E_{gg}(r, r')]$ .
3. *Transverse momenta in the numerator.* The numerator of (4.82) contains no transverse momenta at all — only colour factors,  $\delta$ 's and plus-momentum ratios. The propagating exchange is a product of two (D.3) vertices, each linear in transverse momenta (Eq. (III.15) below), divided by an intermediate denominator that is quadratic in them.
4. *The paper's own  $s$ -channel.* For the  $s$ -channel the two parts are treated separately: the instantaneous term sits in (4.82) (third colour structure), and the propagating term is (4.84), with its own intermediate state. The  $t$ - and  $u$ -channel analogues of (4.84) are absent from Sec. 4.3.1.

Finally, unitarity requires them (Step 12 below).

### III.1.4 Step-by-step calculation of the $t$ -channel exchange

*Step 0: notation and conventions.* Incoming gluons  $g_i^a(k)$ ,  $g_j^b(p)$ ; outgoing  $g_l^d(r)$ ,  $g_n^c(q)$  (the paper's label  $k$  for the polarisation of  $q$  is renamed  $n$ ); exchanged gluon  $g_m^e(r')$ .  $E_i = E_{gg}(k, p) =$

$E_g(k) + E_g(p)$  and  $E_f = E_{gg}(r, q)$ . The states are normalised as in App. B, so the identity operator in the soft Fock space is

$$1 = |0\rangle\langle 0| + \int d^3u |g(u)\rangle\langle g(u)| + \frac{1}{2!} \int d^3u d^3v |g(u)g(v)\rangle\langle g(u)g(v)| \\ + \frac{1}{3!} \int d^3u d^3v d^3w |g(u)g(v)g(w)\rangle\langle g(u)g(v)g(w)| + \dots, \quad (\text{III.5})$$

with sums over colours and polarisations implied. The three-gluon vertex (D.3) is

$$\langle g_l^c(q) g_j^b(p) | H_{ggg} | g_i^a(k) \rangle = \frac{ig}{8\pi^{3/2}} f^{abc} \frac{\delta^{(3)}(k-p-q)}{\sqrt{k^+ p^+ q^+}} V_{ijl}(k; p, q), \quad (\text{III.6})$$

with  $V_{ijl}$  the bracket of (D.3). It is symmetric under exchange of the two daughters:  $(p, b, j) \leftrightarrow (q, c, l)$  flips the sign of both  $f^{abc}$  and  $V_{ijl}$ . Because  $H$  is Hermitian, the merging matrix element is the complex conjugate:

$$\langle g_i^a(k) | H_{ggg} | g_j^b(p) g_l^c(q) \rangle = -\frac{ig}{8\pi^{3/2}} f^{abc} \frac{\delta^{(3)}(k-p-q)}{\sqrt{k^+ p^+ q^+}} V_{ijl}(k; p, q). \quad (\text{III.7})$$

*Step 1: second-order formula.* The second-order part of the wave function of the incoming state  $|i\rangle = |g(k)g(p)\rangle$  is (Eq. (E.3), or (3.5) iterated)

$$|\Psi^{(2)}\rangle = \sum_f \sum_m \frac{|f\rangle\langle f|H|m\rangle\langle m|H|i\rangle}{(E_i - E_f)(E_i - E_m)}. \quad (\text{III.8})$$

For two-gluon final states and three-gluon intermediate states, inserting (III.5) gives the factor  $\frac{1}{2!} \cdot \frac{1}{3!} = \frac{1}{12}$ , exactly as in the paper's (4.87) and (4.91):

$$|\Psi_{3g}^{(2)}\rangle = \frac{1}{12} \int d^3r d^3q d^3u d^3v d^3w |g(r)g(q)\rangle \\ \times \frac{\langle g(r)g(q) | H_{ggg} | g(u)g(v)g(w) \rangle \langle g(u)g(v)g(w) | H_{ggg} | g(k)g(p) \rangle}{(E_i - E_f)(E_i - E_{ggg}(u, v, w))}. \quad (\text{III.9})$$

*Step 2: the first vertex.* The splitting part of  $H_{ggg}$  removes one incoming gluon and creates two. The other incoming gluon is untouched, so it must be one of  $u, v, w$ :

$$\langle uvw | H_{ggg} | kp \rangle = \sum_{x \in \{u, v, w\}} \left[ \langle g(x) | g(p) \rangle \langle g(y)g(z) | H_{ggg} | g(k) \rangle + \langle g(x) | g(k) \rangle \langle g(y)g(z) | H_{ggg} | g(p) \rangle \right],$$

where  $\{y, z\}$  are the two momenta other than  $x$ . The state  $|g(u)g(v)g(w)\rangle$  is symmetric and  $u, v, w$  are integrated, so the three choices of the spectator  $x$  give equal contributions:  $\frac{1}{3!} \times 3 = \frac{1}{2}$ . Calling the spectator  $w$  and doing its integral,

$$\frac{1}{3!} \int d^3u d^3v d^3w |uvw\rangle \langle uvw | H_{ggg} | kp \rangle = \frac{1}{2} \int d^3u d^3v |g(u)g(v)g(p)\rangle \langle g(u)g(v) | H_{ggg} | g(k) \rangle \\ + \frac{1}{2} \int d^3u d^3v |g(u)g(v)g(k)\rangle \langle g(u)g(v) | H_{ggg} | g(p) \rangle. \quad (\text{III.10})$$

In the first term the intermediate energy is  $E_g(u) + E_g(v) + E_g(p)$ , so the spectator drops out of the intermediate denominator:  $E_i - E_m = E_g(k) - E_g(u) - E_g(v)$ .

*Step 3: the second vertex.* The merging part of  $H_{ggg}$  acts on  $|g(u)g(v)g(p)\rangle$ : two of the three gluons merge into one, and the third is a spectator that must coincide with  $r$  or  $q$ . This gives six terms:

- (i)  $u + v \rightarrow q$  with  $p = r$ , or  $u + v \rightarrow r$  with  $p = q$ : the gluon  $k$  splits and recombines, a self-energy of  $k$  with  $p$  as spectator. This is not an exchange; it belongs to the one-gluon sector ( $\mathbb{B}_3$ ) and the normalisation, and is dropped here.
- (ii)  $u + p \rightarrow q$  with  $v = r$ ;
- (iii)  $v + p \rightarrow q$  with  $u = r$ ;
- (iv)  $u + p \rightarrow r$  with  $v = q$ ;
- (v)  $v + p \rightarrow r$  with  $u = q$ .

Terms (ii) and (iii) are the  $t$ -channel ( $k \rightarrow r$ ); terms (iv) and (v) are the  $u$ -channel ( $k \rightarrow q$ ). Because  $\langle g(u)g(v)|H_{ggg}|g(k)\rangle$  is symmetric under  $u \leftrightarrow v$ , (ii) and (iii) are equal, and so are (iv) and (v). The factor 2 cancels the  $\frac{1}{2}$  of (III.10). The same happens for the second term of (III.10), in which  $p$  splits and  $k$  absorbs.

*Step 4: the result in terms of matrix elements.* Collecting the terms,

$$|\Psi^{t+u}\rangle = \frac{1}{2} \int d^3r d^3q |g_l^d(r) g_n^c(q)\rangle [T(r, q) + T(q, r)] = \int d^3r d^3q |g_l^d(r) g_n^c(q)\rangle T(r, q), \quad (\text{III.11})$$

where  $T(q, r)$  is the  $u$ -channel (obtained from  $T(r, q)$  by  $r \leftrightarrow q$ ,  $d \leftrightarrow c$ ,  $l \leftrightarrow n$ ). The last step uses  $|g(r)g(q)\rangle = |g(q)g(r)\rangle$  and relabels the integration variables. Here  $T = T^{(a)} + T^{(b)}$  with

$$T^{(a)}(r, q) = \frac{1}{E_i - E_f} \int d^3r' \frac{\langle g_n^c(q)|H_{ggg}|g_m^e(r') g_j^b(p)\rangle \langle g_l^d(r) g_m^e(r')|H_{ggg}|g_i^a(k)\rangle}{E_g(k) - E_g(r) - E_g(r')}, \quad (\text{III.12})$$

$$T^{(b)}(r, q) = \frac{1}{E_i - E_f} \int d^3r'' \frac{\langle g_l^d(r)|H_{ggg}|g_m^e(r'') g_i^a(k)\rangle \langle g_n^c(q) g_m^e(r'')|H_{ggg}|g_j^b(p)\rangle}{E_g(p) - E_g(q) - E_g(r'')}. \quad (\text{III.13})$$

In  $T^{(a)}$ ,  $k$  emits  $r'$  and becomes  $r$ , and  $p$  absorbs  $r'$  and becomes  $q$  (Fig. 1(a)). In  $T^{(b)}$ ,  $p$  emits  $r''$  and becomes  $q$ , and  $k$  absorbs it and becomes  $r$  (Fig. 1(b)).

*Step 5: insert the vertices.* For the splitting  $k \rightarrow r' + r$ , use (III.6) with parent  $(a, i, k)$ , “ $p$ -slot”  $(e, m, r')$  and “ $q$ -slot”  $(d, l, r)$ :

$$\langle g_l^d(r) g_m^e(r')|H_{ggg}|g_i^a(k)\rangle = \frac{ig}{8\pi^{3/2}} f^{aed} \frac{\delta^{(3)}(k - r' - r)}{\sqrt{k^+ r'^+ r^+}} V_{iml}(k; r', r).$$

For the merging  $r' + p \rightarrow q$ , use (III.7) with parent  $(c, n, q)$ , “ $p$ -slot”  $(e, m, r')$  and “ $q$ -slot”  $(b, j, p)$ :

$$\langle g_n^c(q)|H_{ggg}|g_m^e(r') g_j^b(p)\rangle = -\frac{ig}{8\pi^{3/2}} f^{ceb} \frac{\delta^{(3)}(q - r' - p)}{\sqrt{q^+ r'^+ p^+}} V_{nmj}(q; r', p).$$

The product of the prefactors is  $(ig)(-ig)/(64\pi^3) = g^2/(64\pi^3)$ . Using the antisymmetry of  $f$ , the colour factor is  $f^{aed} f^{ceb} = (-f^{ead})(-f^{ecb}) = f^{ead} f^{ecb}$ . This is exactly the colour structure  $f^{a'ad} f^{a'cb}$  (with  $a' = e$ ) that multiplies the  $t$ -channel instantaneous term in (4.82), as it must be.

*Step 6: the  $r'$  integral.*  $\delta^{(3)}(k - r' - r)$  sets  $r' = k - r$ . Since  $r'^+ > 0$ ,  $T^{(a)}$  exists only for  $k^+ > r^+$ . The second  $\delta$ -function becomes  $\delta^{(3)}(q - p - k + r) = \delta^{(3)}(k + p - r - q)$ , i.e. overall momentum conservation. The square roots combine to  $\sqrt{k^+ r'^+ r^+} \sqrt{q^+ r'^+ p^+} = r'^+ \sqrt{k^+ p^+ q^+ r^+} = (k^+ - r^+) \sqrt{k^+ p^+ q^+ r^+}$ .

*Step 7: the intermediate energy denominator.* Write  $r^+ = \xi k^+$  with  $0 < \xi < 1$  and  $\mathbf{r} = \xi \mathbf{k} + \boldsymbol{\kappa}$ , so that  $\boldsymbol{\kappa} \equiv \mathbf{r} - \xi \mathbf{k}$  is the transverse momentum of  $r$  relative to the direction of  $k$ . Then  $r'^+ = (1 - \xi)k^+$  and  $\mathbf{r}' = (1 - \xi)\mathbf{k} - \boldsymbol{\kappa}$ , and

$$E_g(r) = \frac{\xi \mathbf{k}^2}{2k^+} + \frac{\mathbf{k} \cdot \boldsymbol{\kappa}}{k^+} + \frac{\boldsymbol{\kappa}^2}{2\xi k^+}, \quad E_g(r') = \frac{(1 - \xi) \mathbf{k}^2}{2k^+} - \frac{\mathbf{k} \cdot \boldsymbol{\kappa}}{k^+} + \frac{\boldsymbol{\kappa}^2}{2(1 - \xi)k^+}.$$

Adding,  $E_g(r) + E_g(r') = E_g(k) + \kappa^2/[2k^+\xi(1-\xi)]$ , so

$$E_g(k) - E_g(r) - E_g(r') = -\frac{\kappa^2}{2k^+\xi(1-\xi)}. \quad (\text{III.14})$$

This is negative and vanishes only for  $\kappa = 0$ , i.e. when  $r$  and  $r'$  are collinear with  $k$ . (This is also the general fact used for Comment 2: a splitting always raises the free light-cone energy.)

*Step 8: the vertex in relative momenta.* The (D.3) bracket simplifies when written in terms of the momentum fraction  $z = p^+/k^+$  of its “ $p$ -slot” gluon and the relative momentum  $\kappa_p = \mathbf{p} - z\mathbf{k}$  (so that  $\mathbf{q} = (1-z)\mathbf{k} - \kappa_p$ ):

$$V_{ijl}(k; p, q) = 2\kappa_p^i \delta_{jl} - \frac{2}{z} \kappa_p^j \delta_{il} - \frac{2}{1-z} \kappa_p^l \delta_{ij}. \quad (\text{III.15})$$

For example, the first bracket of (D.3) is  $p^i - q^i + \frac{q^+ - p^+}{k^+} k^i = (p - zk)^i - (q - (1-z)k)^i = 2\kappa_p^i$ ; the other two follow in the same way. Each vertex is linear in a relative transverse momentum and vanishes in the collinear limit. For the splitting  $k \rightarrow r' + r$  the “ $p$ -slot” gluon is  $r'$ , with  $z = 1 - \xi$  and  $\kappa_{r'} = \mathbf{r}' - (1 - \xi)\mathbf{k} = -\kappa$ :

$$V_{iml}(k; r', r) = -2\kappa^i \delta_{ml} + \frac{2}{1-\xi} \kappa^m \delta_{il} + \frac{2}{\xi} \kappa^l \delta_{im}. \quad (\text{III.16})$$

For the merging  $r' + p \rightarrow q$ , write  $p^+ = \eta q^+$  and  $\lambda \equiv \mathbf{p} - \eta\mathbf{q}$  (the relative momentum of the pair  $(p, r')$  that forms  $q$ ). The “ $p$ -slot” gluon is  $r'$ , with  $z = 1 - \eta$  and  $\kappa_{r'} = -\lambda$ :

$$V_{nmj}(q; r', p) = -2\lambda^n \delta_{mj} + \frac{2}{1-\eta} \lambda^m \delta_{nj} + \frac{2}{\eta} \lambda^j \delta_{nm}. \quad (\text{III.17})$$

*Step 9: the result.* Putting Steps 5–8 together, the factor  $1/\{(k^+ - r^+)[E_g(k) - E_g(r) - E_g(r')]\} = -2k^+\xi(1-\xi)/[k^+(1-\xi)\kappa^2] = -2\xi/\kappa^2$ , and

$$T^{(a)}(r, q) = -\frac{g^2}{32\pi^3} f^{ead} f^{ecb} \frac{\delta^{(3)}(k + p - r - q) \Theta(k^+ - r^+)}{\sqrt{k^+ p^+ q^+ r^+}} \frac{\xi P_{ilnj}}{\kappa^2 [E_{gg}(k, p) - E_{gg}(r, q)]}, \quad (\text{III.18})$$

with the polarisation tensor  $P_{ilnj} \equiv \sum_m V_{iml}(k; r', r) V_{nmj}(q; r', p)$ , which from (III.16), (III.17) is

$$\begin{aligned} P_{ilnj} = & 4\kappa^i \lambda^n \delta_{lj} - \frac{4}{1-\eta} \kappa^i \lambda^l \delta_{nj} - \frac{4}{\eta} \kappa^i \lambda^j \delta_{nl} - \frac{4}{1-\xi} \kappa^j \lambda^n \delta_{il} + \frac{4(\kappa \cdot \lambda)}{(1-\xi)(1-\eta)} \delta_{il} \delta_{nj} \\ & + \frac{4}{(1-\xi)\eta} \kappa^n \lambda^j \delta_{il} - \frac{4}{\xi} \kappa^l \lambda^n \delta_{ij} + \frac{4}{\xi(1-\eta)} \kappa^l \lambda^i \delta_{nj} + \frac{4}{\xi\eta} \kappa^l \lambda^j \delta_{in}. \end{aligned} \quad (\text{III.19})$$

Behaviour:  $P$  is linear in  $\kappa$ , so  $T^{(a)} \sim 1/|\kappa|$  as  $\kappa \rightarrow 0$ . The intermediate denominator therefore produces only an integrable singularity at the collinear endpoint, as in  $\mathbb{C}$ . The only non-integrable issue is the overall denominator  $E_{gg}(k, p) - E_{gg}(r, q)$ , which is Comment 2.

*Step 10: the other ordering.*  $T^{(b)}$  follows from  $T^{(a)}$  by the relabelling  $(k, a, i) \leftrightarrow (p, b, j)$ ,  $(r, d, l) \leftrightarrow (q, c, n)$ : “ $k$  emits and becomes  $r$ ,  $p$  absorbs and becomes  $q$ ” turns into “ $p$  emits and becomes  $q$ ,  $k$  absorbs and becomes  $r$ ”. The colour factor is invariant,  $f^{ebc} f^{eda} = f^{ead} f^{ecb}$ . Hence  $T^{(b)}$  is (III.18) with  $\xi \rightarrow \tilde{\xi} = q^+/p^+$ ,  $\kappa \rightarrow \tilde{\kappa} = \mathbf{q} - \tilde{\xi}\mathbf{p}$ ,  $\eta \rightarrow \tilde{\eta} = k^+/r^+$ ,  $\lambda \rightarrow \tilde{\lambda} = \mathbf{k} - \tilde{\eta}\mathbf{r}$ , and  $\Theta(k^+ - r^+) \rightarrow \Theta(r^+ - k^+)$ . Exactly one of  $T^{(a)}$ ,  $T^{(b)}$  is non-zero for given external momenta.

*Step 11: the coefficient of  $\Omega$ .* In the conventions of (4.17), the two-gluon component of  $\Omega[g(k)g(p)]$  is  $\int \frac{d^3 q}{(2\pi)^3} \frac{d^3 r}{(2\pi)^3} [\mathbb{D}_2^{dcba}(k, p, q, r) + \mathbb{D}_2^{dcab}(p, k, q, r)] |g(q)g(r)\rangle$ . Compare with (III.11) in the form  $\frac{1}{2} \iint |rq\rangle [T(r, q) + T(q, r)]$ .  $T(q, r)$  is the  $(k, a, i) \leftrightarrow (p, b, j)$  image of  $T(r, q)$ : under this swap, ordering (a) of the  $t$ -channel goes into ordering (b) of the  $u$ -channel and vice versa. Hence

$$\begin{aligned} \mathbb{D}_{2lnji}^{(t)dcba}(k, p, q, r) &= \frac{1}{2}(2\pi)^6 T(r, q) \\ &= -\pi^3 g^2 f^{ead} f^{ecb} \frac{\delta^{(3)}(k + p - r - q)}{\sqrt{k^+ p^+ q^+ r^+}} \frac{\Theta(k^+ - r^+) \xi P/\kappa^2 + \Theta(r^+ - k^+) \tilde{\xi} \tilde{P}/\tilde{\kappa}^2}{E_{gg}(k, p) - E_{gg}(r, q)}. \end{aligned} \quad (\text{III.20})$$

In terms of  $\mathbb{C}$ : Eq. (5.5) says  $\mathbb{C} = (2\pi)^{9/2} \langle gg | H_{ggg} | g \rangle / [2(E_{\text{parent}} - E_{\text{daughters}})]$ , so

$$\begin{aligned} \langle g(r)g(r') | H_{ggg} | g(k) \rangle &= 2(2\pi)^{-9/2} [E_g(k) - E_g(r) - E_g(r')] \mathbb{C}_{lmi}^{dea}(k, r', r), \\ \langle g(q) | H_{ggg} | g(r')g(p) \rangle &= 2(2\pi)^{-9/2} [E_g(q) - E_g(r') - E_g(p)] (\mathbb{C}_{jmn}^{bec}(q, r', p))^*. \end{aligned}$$

Inserting these in (III.12), the intermediate denominator cancels, and

$$\mathbb{D}_2^{(t)dcba}(k, p, q, r) = 2 \int \frac{d^3 r'}{(2\pi)^3} \mathbb{C}_{jmn}^{bec\dagger}(q, r', p) \mathbb{C}_{lmi}^{dea}(k, r', r) \frac{E_g(q) - E_g(r') - E_g(p)}{E_{gg}(k, p) - E_{gg}(r, q)} + (\text{ordering (b)}). \quad (\text{III.21})$$

*Remark on the factor 2.* The  $s$ -channel term of  $\mathbb{D}_2$  has coefficient 1 instead of 2. This is bookkeeping, not physics. The  $s$ -channel amplitude  $A_s(r, q)$  is already symmetric under  $r \leftrightarrow q$ , because the vertex  $\langle g(r)g(q) | H_{ggg} | g(r') \rangle$  contains both assignments, and it is shared equally between  $\mathbb{D}_2(k, p, q, r)$  and  $\mathbb{D}_2(p, k, q, r)$ . The  $t$ - and  $u$ -channel amplitudes are separate functions,  $T(r, q)$  and  $T(q, r)$ , each assigned to one of the two  $\mathbb{D}_2$ 's.

*Step 12: the unitarity check.* Take the intermediate state  $m = |g(r)g(r')g(p)\rangle$  of Fig. 1(a). Then  $\omega_1(m \leftarrow i)$  is the splitting  $k \rightarrow r' + r$  with  $p$  as spectator, i.e.  $\mathbb{C}(k; r', r)$ , and  $\omega_1(m \leftarrow f)$  is the splitting  $q \rightarrow r' + p$  with  $r$  as spectator, i.e.  $\mathbb{C}(q; r', p)$ . The amplitude (III.21) is the product of the first-order amplitudes  $\omega_1(f \leftarrow m) = -\mathbb{C}^\dagger(q; r', p)$  and  $\omega_1(m \leftarrow i) = \mathbb{C}(k; r', r)$  with the weight  $(E_m - E_f)/(E_i - E_f)$ ,  $E_m - E_f = E_g(r') + E_g(p) - E_g(q)$ , as in Eq. (4). The reversed process  $r q \rightarrow k p$  through the same intermediate state has amplitude  $[-\mathbb{C}^\dagger(k; r', r)][\mathbb{C}(q; r', p)](E_m - E_i)/(E_f - E_i)$ . Taking its Hermitian conjugate and adding (the weights are real):

$$\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger = -\mathbb{C}^\dagger(q; r', p) \mathbb{C}(k; r', r) \left[ \frac{E_m - E_f}{E_i - E_f} + \frac{E_m - E_i}{E_f - E_i} \right] = -\mathbb{C}^\dagger(q; r', p) \mathbb{C}(k; r', r), \quad (\text{III.22})$$

because the bracket equals  $(E_m - E_f - E_m + E_i)/(E_i - E_f) = 1$ . The right-hand side is a  $\mathbb{C}^\dagger \mathbb{C}$  product in which the internal momentum  $r'$  joins the incoming line  $k$  to the outgoing line  $q$  — the “ $t$ -type” terms on the right-hand side of (4.11). Without  $\mathbb{D}_2^{(t)}$  the left-hand side has nothing to match them, so (4.11) cannot hold. With  $\mathbb{D}_2^{(t)}$  it holds identically, and the overall pole has cancelled.

*Step 13: cross-check with the  $s$ -channel term of (5.7).* The same steps applied to (4.84) (intermediate state  $|g(r')\rangle$ ,  $r' = k + p$ ) give

$$\begin{aligned} A_s &= \int d^3 r' \frac{\langle g(r)g(q) | H_{ggg} | g(r') \rangle \langle g(r') | H_{ggg} | g(k)g(p) \rangle}{(E_i - E_f)(E_i - E_g(r'))} \\ &= 4(2\pi)^{-9} \int d^3 r' \mathbb{C}(r'; q, r) \mathbb{C}^\dagger(r'; k, p) \frac{E_g(r) + E_g(q) - E_g(r')}{E_i - E_f}, \end{aligned}$$

where we used  $\langle g(r') | H_{ggg} | g(k)g(p) \rangle = 2(2\pi)^{-9/2} [E_g(r') - E_g(k) - E_g(p)] \mathbb{C}^\dagger(r'; k, p)$  and  $[E_g(r') - E_g(k) - E_g(p)]/[E_i - E_g(r')] = -1$ . With the final-state factor  $\frac{1}{2}$  and the equal sharing between the two  $\mathbb{D}_2$ 's,  $\mathbb{D}_2^{(s)} = \frac{1}{4}(2\pi)^6 A_s$ :

$$\mathbb{D}_2^{(s)}(k, p, q, r) = \int \frac{d^3 r'}{(2\pi)^3} \mathbb{C}(r'; q, r) \mathbb{C}^\dagger(r'; k, p) \frac{E_g(r) + E_g(q) - E_g(r')}{E_{gg}(k, p) - E_{gg}(r, q)}. \quad (\text{III.23})$$

The printed  $s$ -channel term of (5.7) is

$$\int \frac{d^3 r'}{(2\pi)^3} \mathbb{C}(r', k, p) \mathbb{C}^\dagger(r', r, q) \frac{E_g(r') - E_g(k) - E_g(p)}{E_{gg}(r, q) - E_{gg}(k, p)}.$$

For these  $\rho$ -independent coefficients,  $\mathbb{C}$  is  $i$  times a real function, so  $\mathbb{C}(r'; k, p) \mathbb{C}^\dagger(r'; r, q) = \mathbb{C}(r'; q, r) \mathbb{C}^\dagger(r'; k, p)$ . The two expressions therefore differ only in the energy weight, and the weights differ by exactly 1:

$$\frac{E_g(r') - E_g(k) - E_g(p)}{E_f - E_i} - \frac{E_g(r) + E_g(q) - E_g(r')}{E_i - E_f} = \frac{E_i - E_f}{E_i - E_f} = 1.$$

So the printed term equals (III.23) plus the pole-free product  $\int \frac{d^3 r'}{(2\pi)^3} \mathbb{C} \mathbb{C}^\dagger$ . The printed weight is the one of the reversed process  $rq \rightarrow kp$ . The  $\mathbb{A}\mathbb{C}$  terms of (5.7) do use the weight  $(E_m - E_f)/(E_i - E_f)$  of Eq. (4): for “absorb  $k$  first, then split  $p$ ” the weight is  $E_g(p) - E_g(q) - E_g(r)$ , and for “split  $p$  first, then absorb  $k$ ” it is  $E_g(k)$ , exactly as printed. The  $s$ -channel term therefore looks like a slip. Please check it against your notes; the extra term would also spoil the unitarity check (III.22) for the  $s$ -channel intermediate state.

## III.2 Comment 2: the vanishing energy denominator

### III.2.1 Where the energy denominators come from

Write light-cone time as  $t \equiv x^+$  and  $H = H_0 + V$ . Following (3.7),  $\Omega$  is the evolution from  $t = -\infty$  to  $t = 0$  in which the interaction is switched on slowly:

$$\Omega_\varepsilon = T \exp\left(-i \int_{-\infty}^0 dt e^{\varepsilon t} V_I(t)\right), \quad V_I(t) = e^{iH_0 t} V e^{-iH_0 t}, \quad \varepsilon \rightarrow 0^+. \quad (\text{III.24})$$

$V_I(t)$  has matrix elements  $\langle f|V_I(t)|i\rangle = e^{i(E_f - E_i)t} V_{fi}$ . Expanding the time-ordered exponential:

*First order.*

$$\langle f|\Omega|i\rangle^{(1)} = -i V_{fi} \int_{-\infty}^0 dt e^{[\varepsilon + i(E_f - E_i)]t} = \frac{-i V_{fi}}{\varepsilon + i(E_f - E_i)} = \frac{V_{fi}}{E_i - E_f + i\varepsilon}. \quad (\text{III.25})$$

*Second order.* With  $t_1 > t_2$  (the later interaction stands on the left),

$$\langle f|\Omega|i\rangle^{(2)} = (-i)^2 \sum_m V_{fm} V_{mi} \int_{-\infty}^0 dt_1 e^{[\varepsilon + i(E_f - E_m)]t_1} \int_{-\infty}^{t_1} dt_2 e^{[\varepsilon + i(E_m - E_i)]t_2}.$$

The  $t_2$  integral gives  $e^{[\varepsilon + i(E_m - E_i)]t_1} / [\varepsilon + i(E_m - E_i)]$ . The remaining  $t_1$  integral is

$$\int_{-\infty}^0 dt_1 e^{[2\varepsilon + i(E_f - E_i)]t_1} = \frac{1}{2\varepsilon + i(E_f - E_i)}.$$

Multiplying by  $(-i)^2 = -1$  and using  $i \cdot i = -1$  in the denominators,

$$\langle f|\Omega|i\rangle^{(2)} = \sum_m \frac{V_{fm} V_{mi}}{(E_i - E_f + 2i\varepsilon)(E_i - E_m + i\varepsilon)}. \quad (\text{III.26})$$

Three remarks: (1) for  $\varepsilon \rightarrow 0$  these are exactly the LCPT formulas (3.5), (E.4) used in the paper; (2) the later vertex stands on the left, which is the operator ordering of the  $\rho$ 's used throughout the paper; (3) at  $n$ -th order the overall denominator carries  $n\varepsilon$ , because the switching factors  $e^{\varepsilon t}$  of the  $n$  vertices add up.

### III.2.2 Why the overall denominator can vanish

Consider two gluons with total plus momentum  $P^+$  and total transverse momentum  $\mathbf{P}$ . Write  $k^+ = \xi P^+$ ,  $p^+ = (1 - \xi)P^+$ ,  $\mathbf{k} = \xi \mathbf{P} + \boldsymbol{\kappa}$ ,  $\mathbf{p} = (1 - \xi)\mathbf{P} - \boldsymbol{\kappa}$ ; equivalently  $\boldsymbol{\kappa} = (1 - \xi)\mathbf{k} - \xi \mathbf{p}$ . Then

$$E_{gg}(k, p) = \frac{(\xi \mathbf{P} + \boldsymbol{\kappa})^2}{2\xi P^+} + \frac{((1 - \xi)\mathbf{P} - \boldsymbol{\kappa})^2}{2(1 - \xi)P^+} = \frac{\mathbf{P}^2}{2P^+} + \frac{\boldsymbol{\kappa}^2}{2P^+ \xi(1 - \xi)}. \quad (\text{III.27})$$

(The cross terms  $\pm \mathbf{P} \cdot \boldsymbol{\kappa} / P^+$  cancel.) The  $\delta$ -function  $\delta^{(3)}(k + p - r - q)$  in (4.82) makes the outgoing  $P^+$  and  $\mathbf{P}$  equal to the incoming ones: it conserves the plus and transverse momenta. It does *not*

fix the light-cone energy. In LCPT,  $p^-$  is not conserved at the vertices, and that is precisely why energy denominators appear. Using (III.27) for both pairs, the  $\mathbf{P}^2$  terms cancel and

$$E_i - E_f = \frac{1}{2P^+} \left[ \frac{\boldsymbol{\kappa}^2}{\xi(1-\xi)} - \frac{\boldsymbol{\kappa}'^2}{\xi'(1-\xi')} \right]. \quad (\text{III.28})$$

For given incoming  $(\xi, \boldsymbol{\kappa})$  this vanishes whenever  $|\boldsymbol{\kappa}'|^2 = |\boldsymbol{\kappa}|^2 \xi'(1-\xi')/[\xi(1-\xi)]$ . For every  $\xi'$  this is a circle in the  $\boldsymbol{\kappa}'$  plane, so the zeros form a two-dimensional surface in the three-dimensional space of outgoing momenta (Fig. 2(c)). The surface contains the forward point  $(\xi', \boldsymbol{\kappa}') = (\xi, \boldsymbol{\kappa})$ , i.e.  $r = k$ ,  $q = p$ , and the exchanged point  $(1-\xi, -\boldsymbol{\kappa})$ , i.e.  $r = p$ ,  $q = k$ . Directly: once the  $\delta$ -function has set  $q = k + p - r$ ,

$$D(r) = E_g(k) + E_g(p) - E_g(r) - E_g(k + p - r), \quad D(p) = E_g(k) + E_g(p) - E_g(p) - E_g(k) = 0.$$

*Numerical example.* Take  $k^+ = p^+ = 1$ ,  $\mathbf{k} = (1, 0)$ ,  $\mathbf{p} = (-1, 0)$ , so  $E_i = 1$ , and impose  $r = k + p - q$  exactly:

| $q = (q^+; q_1, q_2)$ | $r = k + p - q$   | $D = E_i - E_f$                 |
|-----------------------|-------------------|---------------------------------|
| (1; 1, 0)             | (1; -1, 0)        | 0 (forward: $q = k$ , $r = p$ ) |
| (1; 0, 1)             | (1; 0, -1)        | 0 (90° scattering)              |
| (1; 0.5, 0.5)         | (1; -0.5, -0.5)   | +0.50                           |
| (1; 0.6, 0.6)         | (1; -0.6, -0.6)   | +0.28                           |
| (1; 0.8, 0.8)         | (1; -0.8, -0.8)   | -0.28                           |
| (0.5; 0.2, 0.1)       | (1.5; -0.2, -0.1) | +0.93                           |

In every row  $r + q = k + p$ , so the  $\delta$ -function is satisfied.  $D$  changes sign between the third and fifth rows, so the zero is crossed, not merely touched. By contrast, for the vacuum incoming state of Sec. 4.1,  $E_i = 0 < E_f$  for every final state, so the denominators there never vanish.

### III.2.3 Which coefficients keep the pole

A vanishing denominator in a single time-ordered diagram does not always produce a pole in the coefficient. Suppose the transition consists of two independent sub-processes (e.g. one gluon splits while another is absorbed by the valence) that change the free energy by  $a$  and  $b$ , so that  $E_i - E_f = a + b$ . Both time orderings occur; they have intermediate denominators  $a$  and  $b$  respectively, and

$$\frac{V_1 V_2}{(a+b)a} + \frac{V_2 V_1}{(a+b)b} = V_1 V_2 \frac{a+b}{(a+b)ab} = \frac{V_1 V_2}{ab} \quad (\text{if } V_1 V_2 = V_2 V_1). \quad (\text{III.29})$$

The overall denominator  $a + b$  has disappeared: the sum is just the product of the two first-order amplitudes. If  $a$  and  $b$  have the same sign, as for two splittings in (4.94),  $a + b \neq 0$  anyway. If they have opposite signs — a splitting and a merging in (4.97)+(4.99), an absorption and a splitting in (4.86)+(4.88), a merging and an emission in (4.90)+(4.92) — then  $a + b$  vanishes in each diagram separately, but the pole cancels in the sum. In (5.7)–(5.9) these pieces appear as products  $\mathbb{A}\mathbb{C}$ ,  $\mathbb{C}^\dagger \mathbb{A}$ ,  $\mathbb{C}\mathbb{C}$ ,  $\mathbb{C}\mathbb{C}^\dagger$  whose energy weights add up to  $E_i - E_f$ , which is the same cancellation.

The condition  $V_1 V_2 = V_2 V_1$  matters for  $\rho$ -dependent amplitudes. For (4.50)+(4.52) (absorption of  $k$  by the valence and emission of  $p$ ) the two orderings come with  $\rho^b(-\mathbf{p})\rho^a(\mathbf{k})$  and  $\rho^a(\mathbf{k})\rho^b(-\mathbf{p})$ , which do not commute. Only the symmetric part  $\{\rho, \rho\}$  is pole-free. Compare (4.54), which has no factor  $1/(p^+ \mathbf{k}^2 - k^+ \mathbf{p}^2)$ , with the commutator term (4.55), which has it.

The pole therefore survives only for connected transitions: in  $\mathbb{B}_3$  (the  $\rho$ -linear terms (4.43), (4.44), (4.46), (4.48) and the commutator term (4.55)) and in the  $\rho$ -independent part of  $\mathbb{D}_2$  ((4.82), (4.84), and the  $t$ - and  $u$ -channels). This is the table in Part I.

### III.2.4 Anatomy of $\omega_2$ : where the pole sits and why it cannot cancel

For a transition  $f \neq i$  set  $x \equiv E_i - E_f$  and  $a \equiv E_i - E_m$  (and  $\varepsilon = 0$ ). The complete  $\mathcal{O}(g^2)$  coefficient is

$$\omega_2(f \leftarrow i) = \frac{V_{fi}^{(2)}}{x} + \sum_m \frac{V_{fm} V_{mi}}{x a}.$$

For the reversed transition,  $x \rightarrow -x$  and  $a \rightarrow E_f - E_m = a - x$ , and Hermiticity gives  $\omega_2(i \leftarrow f)^\dagger = V_{fi}^{(2)}/(-x) + \sum_m V_{fm} V_{mi}/[(-x)(a - x)]$ . Taking the sum and the difference:

$$\begin{aligned} \frac{1}{2} [\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger] &= \frac{1}{2} \sum_m V_{fm} V_{mi} \frac{(a - x) - a}{x a (a - x)} = -\frac{1}{2} \sum_m \frac{V_{fm} V_{mi}}{(E_i - E_m)(E_f - E_m)} \\ &= -\frac{1}{2} \sum_m \omega_1(m \leftarrow f)^\dagger \omega_1(m \leftarrow i), \end{aligned} \quad (\text{III.30})$$

$$\frac{1}{2} [\omega_2(f \leftarrow i) - \omega_2(i \leftarrow f)^\dagger] = \frac{1}{E_i - E_f} \left[ V_{fi}^{(2)} + \frac{1}{2} \sum_m V_{fm} V_{mi} \left( \frac{1}{E_i - E_m} + \frac{1}{E_f - E_m} \right) \right]. \quad (\text{III.31})$$

(For the second line:  $\frac{1}{xa} + \frac{1}{2a(a-x)} = \frac{2(a-x)+x}{2xa(a-x)} = \frac{1}{2x} \left[ \frac{1}{a} + \frac{1}{a-x} \right]$ .) Three conclusions:

1. The Hermitian part (III.30) has no pole, and it is completely fixed by unitarity: it is the  $\mathcal{O}(g^2)$  condition (4.5).
2. The pole sits entirely in the anti-Hermitian part (III.31), which is the  $\mathcal{O}(g^2)$  part of  $i\mathbb{G}$ , the generator of (5.11).
3. On the energy shell,  $E_f = E_i$ , the residue of the pole is

$$T_{fi} = V_{fi}^{(2)} + \sum_m \frac{V_{fm} V_{mi}}{E_i - E_m}, \quad (\text{III.32})$$

which is the second-order (Born) on-shell scattering amplitude. For  $\mathbb{D}_2$  it is the tree-level  $gg \rightarrow gg$  amplitude: contact,  $s$ -,  $t$ - and  $u$ -channel exchange, each with its instantaneous part. For  $\mathbb{B}_3$  it is the amplitude for a soft gluon to scatter off the colour field of the valence charges. These are physical amplitudes and are non-zero in general, so the pole is genuine and cannot cancel. Incidentally, only the sum of all four exchanges gives the gauge-invariant on-shell amplitude, which is another reason why the  $t$ - and  $u$ -channel terms of Comment 1 are needed.

### III.2.5 Without a prescription (the basis of Version B)

(a) *What goes wrong.* A function with a simple pole on a surface cannot be integrated across it in the ordinary sense, since  $\int |dx/x|$  diverges. The value of  $\int_{-a}^b dx/x$  (with  $a, b > 0$ ) depends on how the pole is approached:

$$\int_{-a}^{-\delta} \frac{dx}{x} + \int_{\delta}^b \frac{dx}{x} = \log \frac{\delta}{a} + \log \frac{b}{\delta} = \log \frac{b}{a}, \quad \int_{-a}^{-\delta} \frac{dx}{x} + \int_{2\delta}^b \frac{dx}{x} = \log \frac{\delta}{a} + \log \frac{b}{2\delta} = \log \frac{b}{a} - \log 2,$$

and any other answer can be produced in the same way. A coefficient with this pole, integrated across the energy shell, therefore has no unique value until a rule is specified.

(b) *Where the paper is safe.* In the paper,  $\mathbb{B}_3$  and  $\mathbb{D}_2$  are never integrated across the shell on their own:

- The normalisation conditions (4.67)–(4.80) and the unitarity relations (4.9)–(4.14) involve only the Hermitian parts  $\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger$ , which are pole-free by (III.30).

- In the diagonalisation of Sec. 6, the  $\mathcal{O}(g^2)$  coefficients enter  $\Omega^\dagger H \Omega$  through  $H_0 \Omega^{(2)} + \Omega^{(2)\dagger} H_0$ , whose matrix elements are

$$E_f \omega_2(f \leftarrow i) + E_i \omega_2(i \leftarrow f)^\dagger = -(E_i - E_f) \omega_2(f \leftarrow i) + E_i [\omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger].$$

The first term has the pole multiplied by its own zero; the second is the Hermitian part. Carrying this through for all  $\mathcal{O}(g^2)$  terms of  $\Omega^\dagger H \Omega = (1 + \omega_1^\dagger + \omega_2^\dagger)(H_0 + V + V^{(2)})(1 + \omega_1 + \omega_2)$ , which are  $V^{(2)} + H_0 \omega_2 + \omega_2^\dagger H_0 + V \omega_1 + \omega_1^\dagger V + \omega_1^\dagger H_0 \omega_1$ , and setting  $E_i = 0$  for brevity ( $E_f = -x$ ,  $E_m = -a$ ), the off-diagonal element is

$$\begin{aligned} & V_{fi}^{(2)} \left(1 + \frac{E_f - E_i}{x}\right) + \sum_m V_{fm} V_{mi} \left[ \frac{E_f}{x a} - \frac{E_i}{x(a-x)} + \frac{1}{a} + \frac{1}{a-x} + \frac{E_m}{a(a-x)} \right] \\ &= V_{fi}^{(2)} (1 - 1) + \sum_m V_{fm} V_{mi} \left[ -\frac{1}{a} + \frac{1}{a} + \frac{1}{a-x} - \frac{1}{a-x} \right] = 0 \end{aligned} \quad (\text{III.33})$$

for every pair of states with  $E_i \neq E_f$ . (In the bracket, the last three terms come from  $V \omega_1$ ,  $\omega_1^\dagger V$  and  $\omega_1^\dagger H_0 \omega_1$ .) So the cancellation of the off-diagonal elements in (6.4)–(6.10) needs no regularisation; only the exactly degenerate pairs, a set of measure zero, are left undetermined.

(c) *Where it is not safe.* Consider an observable  $\langle \Psi | O | \Psi \rangle$  with an operator  $O$  that does not commute with  $H_0$ , for example the gluon number at fixed momentum after the eikonal scattering off the target, as in [44]. Its  $\mathcal{O}(g^2)$  term contains  $(O_f - O_i) \omega_2(f \leftarrow i)$ , and this pole is not cancelled. Such integrals need a prescription. This is why Version B has to postpone the issue to [44], and why the referee may not accept it.

### III.2.6 With the prescription (the basis of Version A)

(a) *Sokhotski–Plemelj.* For real  $x$  and  $\eta > 0$ ,

$$\frac{1}{x + i\eta} = \frac{x}{x^2 + \eta^2} - \frac{i\eta}{x^2 + \eta^2}.$$

As  $\eta \rightarrow 0$  the first term tends to the principal value  $\text{P}(1/x)$  (the symmetric prescription of III.2.5(a)). The second is a Lorentzian of area  $\pi$ , since  $\int dx \eta / (x^2 + \eta^2) = \pi$ , and it tends to  $\pi \delta(x)$ . Hence

$$\frac{1}{x + i0} = \text{P} \frac{1}{x} - i\pi \delta(x). \quad (\text{III.34})$$

Example:  $\int_{-a}^b \frac{dx}{x + i\eta} = \log \frac{b + i\eta}{-a + i\eta} \rightarrow \log \frac{b}{a} - i\pi$  (numerically, for  $a = 1$ ,  $b = 2$ ,  $\eta = 10^{-4}$ :  $0.693147 - 3.14144i$ ). The answer is now unique.

(b) *Unitarity holds for every  $\varepsilon$ .* First order:  $\omega_1(i \leftarrow f)^\dagger = [V_{if} / (E_f - E_i + i\varepsilon)]^* = V_{fi} / (E_f - E_i - i\varepsilon) = -\omega_1(f \leftarrow i)$ . Second order, with  $x = E_i - E_f$  and  $a = E_i - E_m$ :

$$\begin{aligned} \omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger &= V_{fm} V_{mi} \left[ \frac{1}{(x + 2i\varepsilon)(a + i\varepsilon)} + \frac{1}{(-x - 2i\varepsilon)(a - x - i\varepsilon)} \right] \\ &= \frac{V_{fm} V_{mi}}{x + 2i\varepsilon} \cdot \frac{(a - x - i\varepsilon) - (a + i\varepsilon)}{(a + i\varepsilon)(a - x - i\varepsilon)} \\ &= -\frac{V_{fm} V_{mi}}{(a + i\varepsilon)(a - x - i\varepsilon)} = -\omega_1(m \leftarrow f)^\dagger \omega_1(m \leftarrow i). \end{aligned}$$

The numerator  $-(x + 2i\varepsilon)$  cancels the overall denominator exactly because the second-order term carries  $2i\varepsilon$ . This is why “ $n\varepsilon$  at  $n$ -th order” matters: with  $i\varepsilon$  at second order the identity would fail.

(c) *Diagonal elements and the normalisation  $\mathbb{N}$ .* For  $f = i$  ( $x = 0$ ):

$$\omega_2(i \leftarrow i) = \sum_m \frac{|V_{mi}|^2}{2i\varepsilon(a + i\varepsilon)} = -\frac{1}{2} \sum_m \frac{|V_{mi}|^2}{a^2 + \varepsilon^2} - \frac{i}{2\varepsilon} \sum_m \frac{|V_{mi}|^2 a}{a^2 + \varepsilon^2}.$$

The real part tends to  $-\frac{1}{2} \sum_m |\omega_1(m \leftarrow i)|^2$ : this is the normalisation  $N_2$  of (4.6). The imaginary part diverges like  $1/\varepsilon$ . It equals  $-iE^{(2)}/(2\varepsilon)$ , where  $E^{(2)} = \sum_m |V_{mi}|^2/(E_i - E_m)$  is the second-order energy shift; this is the expansion of the phase  $e^{-iE^{(2)}/(2\varepsilon)}$  accumulated during the adiabatic switching. It is removed by the Gell-Mann–Low normalisation, which is the “overall phase” mentioned after (3.7).

(d) *The transformed Hamiltonian.* Redo (III.33) keeping the  $i\varepsilon$ ’s. At first order,

$$V_{fi} + E_f \omega_1(f \leftarrow i) + E_i \omega_1(i \leftarrow f)^\dagger = V_{fi} \left[ 1 - \frac{E_i - E_f}{E_i - E_f + i\varepsilon} \right] = V_{fi} \frac{i\varepsilon}{E_i - E_f + i\varepsilon}.$$

At second order the same algebra gives the remainder

$$R_{fi} = V_{fi}^{(2)} \frac{i\varepsilon}{x + i\varepsilon} + \sum_m \frac{V_{fm} V_{mi}}{a + i\varepsilon} \cdot \frac{2i\varepsilon}{x + 2i\varepsilon} + \sum_m \frac{i\varepsilon V_{fm} V_{mi}}{(a - x - i\varepsilon)(a + i\varepsilon)}. \quad (\text{III.35})$$

Three cases:

- $E_i \neq E_f$ :  $R_{fi} \rightarrow 0$  as  $\varepsilon \rightarrow 0$ . The off-diagonal elements cancel, as in the paper.
- $E_i = E_f$ ,  $f \neq i$ :  $R_{fi} \rightarrow V_{fi}^{(2)} + \sum_m V_{fm} V_{mi}/(E_i - E_m) = T_{fi}$ , the on-shell amplitude (III.32). This is the coupling between degenerate states that no unitary transformation can remove.
- $f = i$ :  $R_{ii} = E^{(2)}$ , the energy shift, e.g. the  $\rho^2$  term of (6.9) for the vacuum.

Between normalisable states (wave packets  $\varphi$  that are smooth in energy), the degenerate part does not contribute:

$$\int dE_f \varphi(E_f) \frac{i\varepsilon}{E_i - E_f + i\varepsilon} = i\varepsilon \left[ \mathcal{P} \int dE_f \frac{\varphi(E_f)}{E_i - E_f} - i\pi \varphi(E_i) \right] \rightarrow 0 \quad (\varepsilon \rightarrow 0).$$

Hence  $\Omega^\dagger H \Omega = H_0 + (\text{diagonal energy shifts})$  on normalisable states, which is the content of (6.10). With this prescription the diagonalisation of Sec. 6 holds as stated.

(e) *Interpretation.* With the  $i\varepsilon$  of (III.24),  $\Omega$  is the Møller (“in”) operator of scattering theory,  $\Omega = \lim_{t_0 \rightarrow -\infty} e^{iHt_0} e^{-iH_0 t_0}$ , and  $\Omega|n\rangle$  are the Lippmann–Schwinger in-states  $|n^{(+)}\rangle = |n\rangle + (E_n - H_0 + i0)^{-1} V |n^{(+)}\rangle$ . The  $\delta$ -function in (III.34) is the on-shell part, which describes the  $2 \rightarrow 2$  scattering (and the scattering off the background) that takes place during the evolution. It is anti-Hermitian, so it sits in the generator  $\mathbb{G}$  and does not affect unitarity.

### III.2.7 Summary: the two versions side by side

|                    | Version A (with prescription)                                                                                                           | Version B (without prescription)                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| What is claimed    | Coefficients are defined everywhere by the $i\varepsilon$ that follows from (3.7); unitarity holds identically; Sec. 6 holds as stated. | No result of this paper depends on the treatment; $\mathbb{B}_3$ , $\mathbb{D}_2$ are given for $E_i \neq E_f$ ; the prescription is left to [44]. |
| Mathematical basis | III.2.1, III.2.6                                                                                                                        | III.2.4, III.2.5(b)                                                                                                                                |
| Added to the paper | One paragraph; (3.7) rewritten; Eq. (7).                                                                                                | One paragraph.                                                                                                                                     |
| Recalculation      | None.                                                                                                                                   | None.                                                                                                                                              |
| Risk               | Low: answers the referee’s request directly.                                                                                            | Medium: the referee asked “how this is done”, and B defers it.                                                                                     |

My recommendation is Version A. If you like, add the one-line observation of Version B, item (ii), to it: it shows that the diagonalisation does not depend on the choice of prescription, which makes the reply stronger.